

BOAMP Procurement Renewal Linking

Final Technical Report — Phase 1 and Phase 2

Detailed Reference for Data Audit, Preprocessing, Renewal Event Construction, and
Survival Analysis

Internship Project — Gigalis

June 2026

Abstract

This report is the **detailed technical reference** for the internship work. It documents the concrete implementation choices, data audit, preprocessing rules, intermediate diagnostics, linkage logic, method comparison, and survival-modeling outputs in a form that can be checked, reproduced, and handed off. BOAMP notices do not contain an official field linking a call-for-tender (*appel d'offre*) to a later recurrence of the same procurement need. The relationship is therefore inferred from observable signals: buyer identity, contract description, CPV classification, and timing relative to the expected contract end date.

Current result after method comparison. Starting from 1,933 *APPEL-OFFRE* notices over 2015–2024, the pipeline defines 1,210 eligible source contracts. The selected current method is **M0 balanced**, the calibrated composite rule with text similarity ≥ 0.50 , composite score ≥ 0.50 , $W = 6$ months, corrected generic CPV handling, and no margin floor. It produces **269 proxy recurrence events** (22.2%) and 941 censored/no observed-link rows. A new method-comparison experiment tested the current rule (M0), a probabilistic linkage model (M1), and an active-learning-assisted variant (M2). Although M2 balanced is the best alternative candidate in the comparison table, the final recommendation is to **retain M0 balanced as the main method** pending further targeted manual review.

Survival analysis under the selected method. The selected M0 balanced dataset has 1,210 rows and 269 proxy events. The Kaplan–Meier median is **not reached**; fixed-horizon survival is 95.9% at 12 months, 92.3% at 24 months, 83.2% at 48 months, and 75.2% at 60 months. The Cox model has C-index 0.544, and the best parametric AFT model is LogNormalAFT (AIC 3542.5). Operational 12- and 24-month indicators are available for 1,204 contracts under M0 balanced, with median probabilities 0.0198 and 0.0624. The older 665-event output is retained only as the pre-calibration baseline; it is not the current main result.

Method-comparison update — 2026-07-08

The latest validation step is `notebooks/06_linkage_method_comparison_no_ground_truth.ipynb`. It compares the current calibrated rule against literature-supported alternatives without assuming that BOAMP contains real recurrence-chain ground truth. The experiment uses the current candidate-pair table, the synthetic benchmark, negative controls, and mapped manual audit labels where available. Real BOAMP precision and recall are not reported as known; only synthetic precision/recall are directly observable.

Method stages. The reports now distinguish six stages:

1. **Old pre-calibration baseline:** the broad 665-event Sentence-Transformer handoff, retained only for provenance.
2. **M0: current calibrated composite rule:** broad/balanced/strict deterministic rules selected by the synthetic benchmark and real diagnostics.
3. **M1: probabilistic linkage:** a logistic match-probability model trained on synthetic pair labels because Splink and recordlinkage were not available in the current environment.
4. **M2: active-learning-assisted linkage:** the same probability feature space with mapped manual TP/FP accepted-link labels and a targeted review sample.
5. **Final selected method:** M0 balanced remains the main method.
6. **Sensitivity versions:** broad and strict variants are retained as sensitivity specifications, not as replacements for the main result.

Table 1: Method-comparison results on real BOAMP. Real BOAMP precision and recall are not directly observable, so this table reports event counts, negative-control behavior, and mapped manual-audit precision where available.

Method	Variant	Events	Event rate	Neg.-control accept.	Manual audit precision
M0	Broad	490	40.5%	41.6%	25.0%
M0	Balanced	269	22.2%	9.4%	43.8%
M0	Strict	79	6.5%	0.4%	85.7%
M1	Balanced	278	23.0%	9.2%	32.4%
M2	Balanced	256	21.2%	7.8%	34.3%

Table 2: Synthetic benchmark metrics across all scenarios. These are computed on synthetic truth known by construction and do not establish real BOAMP truth.

Method	Variant	Precision	Recall	False positives	False negatives
M0	Broad	0.476	0.645	1,604	802
M0	Balanced	0.575	0.568	949	976
M0	Strict	0.775	0.381	250	1,398
M1	Balanced	0.679	0.642	687	810
M2	Balanced	0.679	0.642	687	810

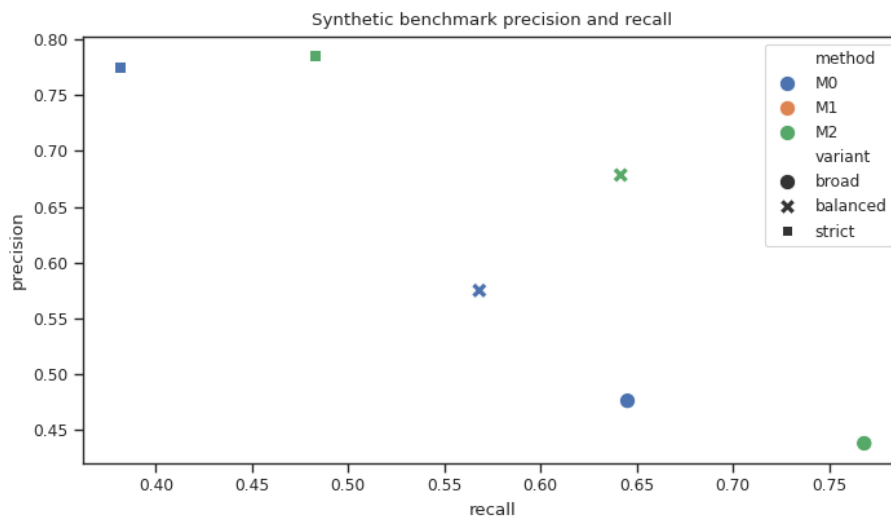


Figure 1: Synthetic benchmark precision and recall for M0, M1, and M2. M1/M2 balanced improve synthetic precision and recall relative to M0 balanced, but the real-data recommendation still retains M0 balanced pending targeted review because real BOAMP labels are proxy recurrences.

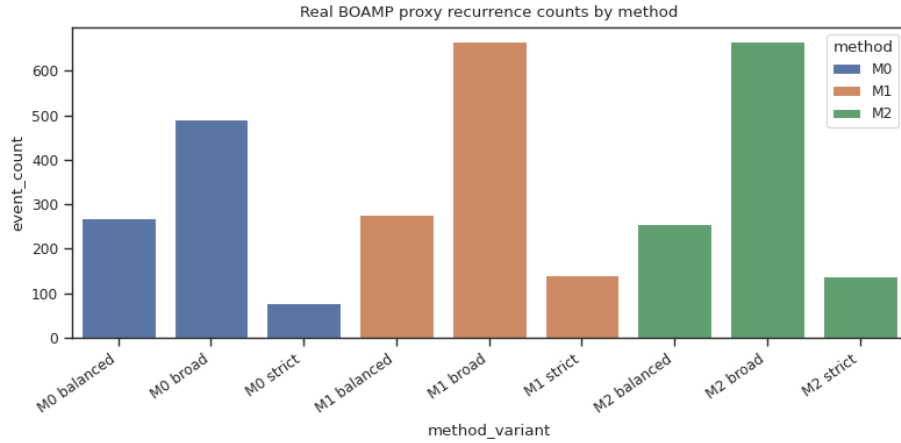


Figure 2: Real BOAMP proxy-event counts by method and variant. The selected main specification remains M0 balanced with 269 events. Broad specifications are upper sensitivity bounds and strict specifications are high-confidence sensitivity bounds.

Final recommendation. The final recommendation table `reports/tables/validation/final_method` selects **M0 balanced** as the current main method. The best alternative balanced method is **M2 balanced**; it has 256 real BOAMP proxy events, a 7.8% negative-control acceptance rate, and a Cox C-index of 0.562 in the required survival comparison. It is kept as a candidate for targeted manual review, not as an immediate replacement.

Survival comparison. The required survival rerun compares M0 balanced, M2 balanced, and M0 strict. For M0 balanced, $n = 1,210$, events = 269, event rate = 22.2%, KM median is not reached, survival is 95.9% at 12 months, 92.3% at 24 months, 83.2% at 48 months, and 75.2% at 60 months. Cox C-index is 0.544 and LogNormalAFT is best by AIC (3,542.5). Because the KM median is not reached, fewer than 50% of contracts experience the calibrated proxy event before censoring.

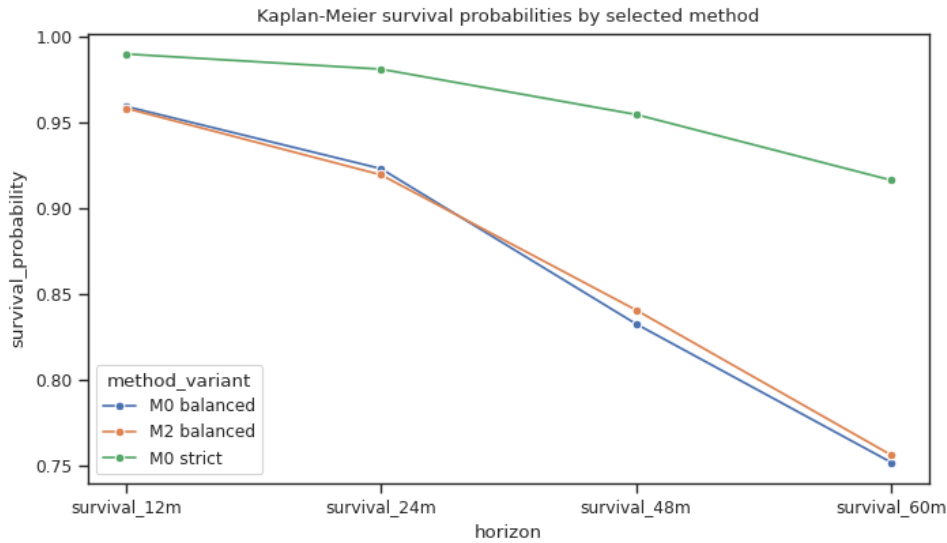


Figure 3: Kaplan–Meier fixed-horizon survival probabilities for the required method comparison: M0 balanced, M2 balanced, and M0 strict.

Active-learning output. The method-comparison notebook writes `reports/tables/validation/active_learning_output.csv` with 150 targeted review pairs: 25 each near the M0 threshold, high-score generic CPV, event=0/high nearest candidate, low top1–top2 margin, high text with CPV mismatch, and same-buyer/same-CPV weak-text cases. This sample is a review queue, not completed validation.

Reproducibility. The executed outputs are `reports/tables/validation/linkage_method_comparison_probabilistic_linkage_results.csv`, `final_method_recommendation.csv`, `method_survival_comparison.csv`, and the figures in `reports/figures/validation/`. The final selected dataset is `data/processed/boamp_phase2_survival_method_m0_balanced.csv`; the older calibrated file `data/processed/boamp_phase2_survival_calibrated_balanced.csv` remains equivalent for the selected M0 balanced event definition.

Calibration update — 2026-07-05

The 665-event output described in the original sections is now treated as the **pre-calibration baseline**, not the recommended input. This section documents why that baseline was replaced and how the replacement was built, validated, and applied.

Initial method and challenge discovered. The original linking rule (Section 6.5) accepted any candidate with a non-negative composite score after a soft text-similarity filter, which maximized the linking rate (665/1,210, 55.0%) but could not be checked against ground truth: BOAMP carries no certified renewal-chain label. A manual audit of 150 stratified cases against the full official BOAMP record (2026-07-02) found raw precision of only

≈ 0.15 (weighted ≈ 0.09 over the 665-event population), rising to 0.50 in the high-confidence tier and to 1.00 only above text similarity 0.80; 12.8% of decided-unlinked sources also hid a plausible missed renewal (Section 14.3). The baseline optimized for linking rate, not for link correctness, and no legal ground truth exists to certify any individual real link. This is the reason the event definition was recalibrated: a synthetic BOAMP benchmark — scored with the same Sentence-Transformer encoder as the real pipeline (`paraphrase-multilingual-MiniLM-L12-v2`) — and real BOAMP diagnostics were used to choose three explicit event-definition rules:

Why a synthetic benchmark? Real BOAMP does not provide certified renewal-chain labels, so the linking task cannot be evaluated on the real data with true precision and recall. The synthetic benchmark was therefore created to provide a controlled evaluation dataset where the true recurrence relation is known by construction, while still reflecting the empirical difficulties seen in BOAMP. Its role is methodological: to test and calibrate the proxy-event definition before using that definition in survival analysis.

How it was built. The synthetic benchmark is fully implemented as a simulation-based BOAMP-like dataset where recurrence truth is known by construction. It generates synthetic source notices, synthetic candidate notices, and an explicit true-link table in three scenarios (easy, medium, hard), with design parameters guided by the real BOAMP data. The simulation intentionally reproduces the main observed failure modes of the linking task: buyer-name variation, generic or drifting CPV codes, missing CPV or duration, short or paraphrased text, timing shifts around the expected end date, and ambiguous large-buyer same-name candidates. The same Sentence-Transformer encoder and the same scoring logic used in the operational pipeline are then applied to the synthetic pairs.

What the benchmark shows. Because the truth is synthetic, the benchmark is not presented as real BOAMP validation; instead it is a controlled stress test that allows direct computation of precision, recall, and F1. Under the final Sentence-Transformer scoring, the balanced rule performs strongly in the easy scenario (precision 0.777, recall 0.806, F1 0.791), remains acceptable in the medium scenario (precision 0.601, recall 0.586, F1 0.593), and deteriorates in the deliberately adverse hard scenario (precision 0.318, recall 0.302, F1 0.310). This result is important substantively because it quantifies both the usefulness and the limits of the linking approach under realistic noise.

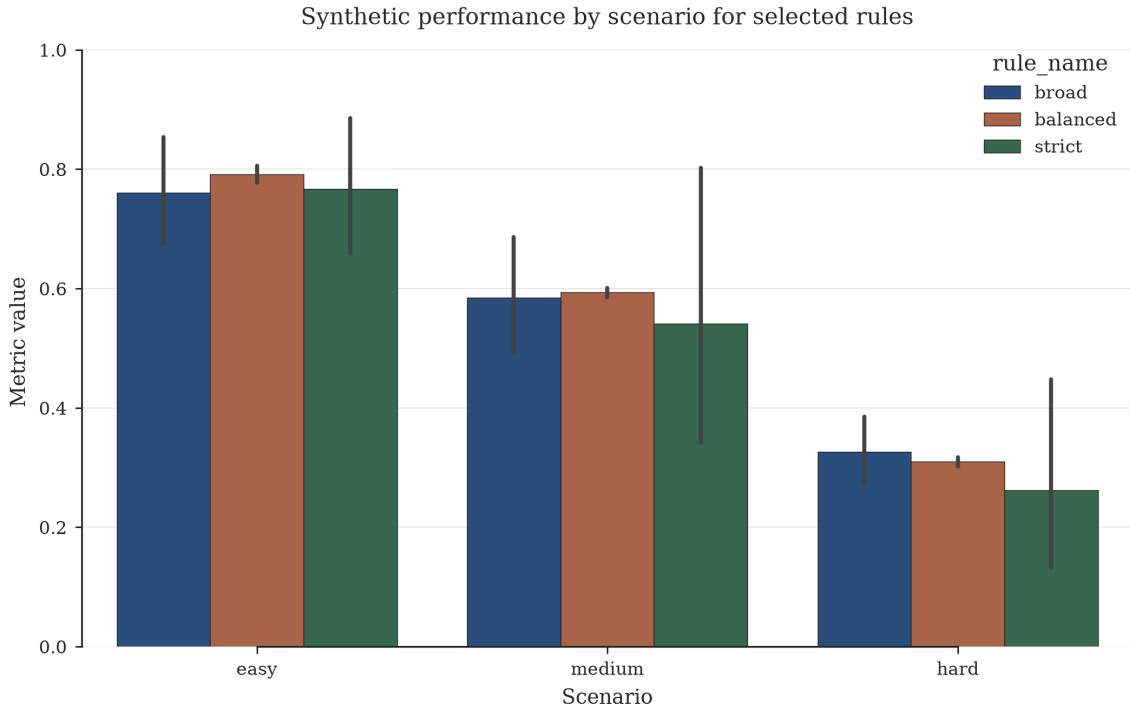


Figure 4: Synthetic-benchmark performance of the broad, balanced, and strict rules across the easy/medium/hard scenarios. All three rules degrade under harder synthetic noise, which is the intended stress test; the balanced rule was retained for its stability across scenarios rather than for being the single most precise option on paper.

How it is used in the pipeline. Operationally, the benchmark is used to score a full threshold grid and then to compare candidate broad, balanced, and strict rules before re-applying them to real BOAMP candidate pairs. The balanced rule was retained because it preserved acceptable synthetic reliability, especially in the medium scenario, while also keeping enough real proxy events for downstream survival modeling. The benchmark therefore acts as the calibration layer between the raw linking algorithm and the final survival dataset: it informs the event-definition rule, while the real BOAMP outputs remain the basis for event counts, Kaplan–Meier estimation, Cox/AFT modeling, and operational risk indicators. The detailed implementation is reproducible from `notebooks/04.synthetic.boamp.benchmark.ipynb` and `notebooks/05.parameter.calibration.benchmark.and.real.ipynb`, with generated artifacts tracked under `data/synthetic/` and `reports/tables/validation/`.

Table 3: Calibrated proxy-event rules

Rule	Text threshold	Composite floor	Margin floor	Events	Use
Broad	0.40	none	none	490	high-recall sensitivity
Balanced	0.50	0.50	none	269	recommended main input
Strict	0.70	0.65	none	79	high-precision sensitivity

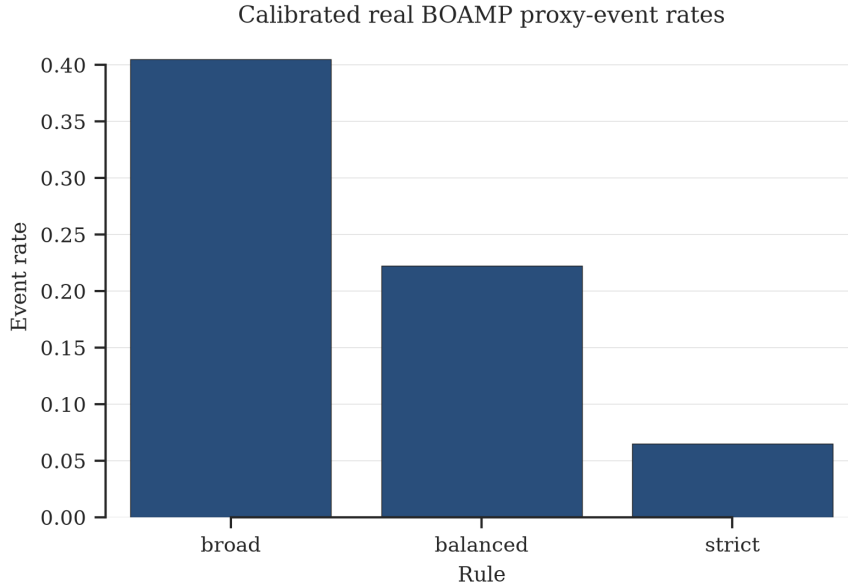


Figure 5: Proxy-event rate on real BOAMP under the three calibrated rules (1,210 eligible contracts). Moving from broad to balanced to strict trades linkage quantity for a rule whose behaviour has been checked on the synthetic benchmark above.

All three use $W = 6$ and the corrected generic-CPV rule, where generic CPV codes do not receive exact-match credit. The recommended survival file is `data/processed/boamp_phase2_survival_cal` (1,210 rows, 269 events, 22.2%). Readiness checks found no missing or non-positive survival durations, no duplicate contract IDs, and no event without a renewal candidate ID; the strict rule is flagged `LOW_EVENTS` (79 events < 100) for Cox/AFT stability.

Under the balanced rule the Kaplan–Meier median is not reached; survival is 95.9% at 12 months, 92.3% at 24 months, 83.2% at 48 months, and 75.2% at 60 months. In the method-comparison survival rerun, the Cox C-index is 0.544, and LogNormalAFT is the best balanced-rule AFT model by AIC (3,542.5, ahead of Weibull 3,570.2 and LogLogistic 3,580.0) — **Weibull was tested but not retained** because it does not have the lowest AIC. Re-scored 12/24-month risk indicators under this rule (Section 15) give mean probabilities of 0.023/0.068 and 27.5 expected recurrences within 12 months. These are diagnostics for identifiable proxy recurrences in BOAMP, not estimates of formally verified renewal-chain probability.

Cox specification and PH assumption check (official M0 balanced spec). The 0.544 C-index above comes from the reduced 3-covariate Cox model (`declared_duration_months`, `dur_was_imputed`, `start_year`; penalizer 0.05) used in the method-comparison rerun so that M0, M1, M2, and the strict variant share identical covariates (`scripts/linkage_method_comparison_no.g`). An earlier, richer model on the same 269-event data adds category-segment dummies (penalizer 0.1) and gives C-index 0.592 (AFT AIC 3,544.8, `notebooks/07_calibrated_survival_analysis.ipynb`); it is kept only as a secondary category-aware fit, not the headline number. The Schoenfeld

residual test on the official 0.544 spec (`reports/tables/survival/m0_balanced_cox_ph_assumption_test`) flags a PH violation for `declared_duration_months` ($\chi^2 = 105.8$, $p < 10^{-24}$) and passes for `dur_was_imputed` ($p = 0.50$) and `start_year` ($p = 0.61$) — the same pattern found in the pre-calibration model earlier in this report. As before, the log-normal AFT model above does not require the PH assumption and is used as the cross-check for individual-level predictions.

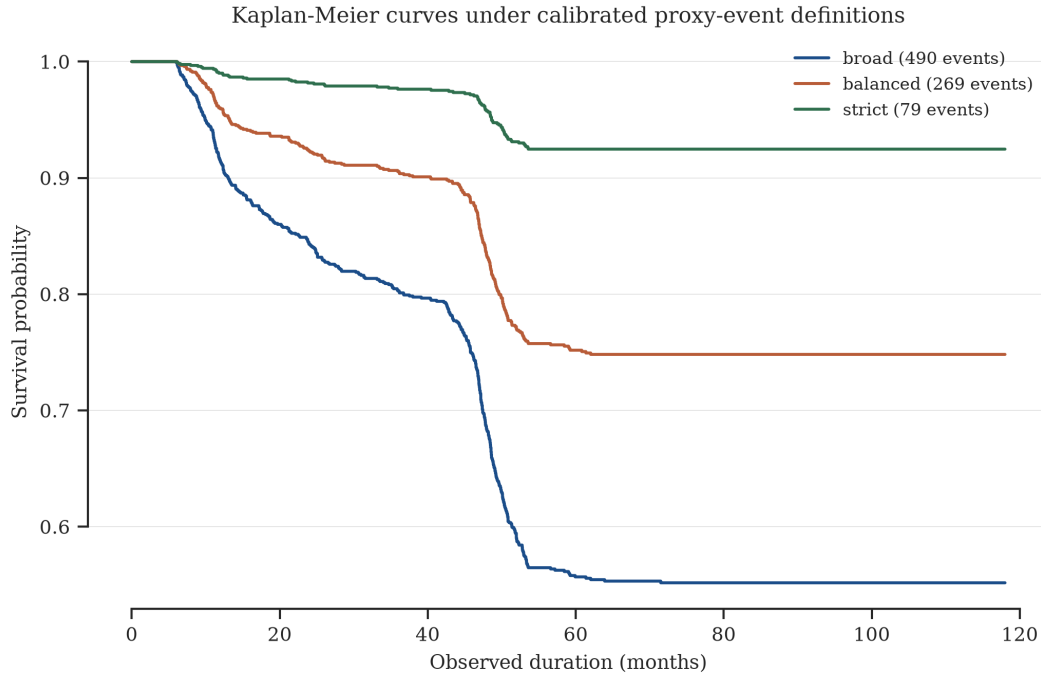


Figure 6: Kaplan–Meier survival curves under the three calibrated proxy-event rules. Stricter rules yield fewer, later-recorded events and higher residual survival at every horizon; the balanced curve sits between the broad and strict extremes, consistent with its intermediate event count.

Limitation. Because BOAMP does not provide an official renewal label, the constructed `event` variable — under any of the three calibrated rules, and under the pre-calibration baseline — should be read as a proxy for likely procurement recurrence, not as verified renewal-chain ground truth. The synthetic benchmark quantifies how the linking method behaves under controlled, known noise; it does not certify that any specific real BOAMP link is a true renewal, and the manual audit above shows precision on real data is far lower than on the easy/medium synthetic scenarios.

Conclusion of the calibration. The calibration phase’s main deliverable is a change of objective, not a single statistic: the pipeline moved from a rule that maximized **linkage quantity** (665 events, 55.0%, later found $\approx 9\text{--}15\%$ precise) to a rule calibrated to maximize **analytical reliability** against a controlled benchmark (269 events, 22.2%). All downstream sections of this report should be read with that distinction in mind: the original sections below

describe the pre-calibration baseline in full mechanical detail (still valid as a description of the algorithm), while this update section gives the numbers to actually use.

How to read this report

This is the **detailed technical reference**. A reader who only wants the scope, workstreams, and headline results should start with the separate **internship report**; the data-reliability and event-definition narrative is in the **data quality report**. This document is the place for the exact mechanics: preprocessing rules, the renewal-linking algorithm, diagnostics, survival-model specifications, and reproducibility. The original detailed sections document the post-SIREN W=6 baseline (665 events, 55.0%); the calibration update above gives the current recommended balanced rule (269 events, 22.2%).

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1 Project Scope

1.1 Business Context

Gigalis manages a portfolio of public digital infrastructure contracts in the Pays-de-la-Loire region. A critical operational question is: *when is a procurement contract likely to be renewed?* Knowing this in advance allows account managers to anticipate renewal campaigns, prepare competitive offers, and reduce the risk of losing contracts by default.

1.2 Why BOAMP?

The **Bulletin Officiel des Annonces de Marchés Publics** (BOAMP) is the official French public procurement gazette. All public contracts above the relevant thresholds must be published there, making it the most complete source of historical procurement data in France. It covers the full lifecycle: pre-information, call for tender (*appel d'offre*), award (*attribution*), and modification notices.

1.3 Core Problem

BOAMP does **not** provide an explicit field linking an original call for tender to its renewal. Each notice is a standalone record. The renewal relationship must therefore be **inferred** from a combination of signals: the same buyer re-issuing a contract for a similar service in the expected time-window after the original contract was due to expire.

1.4 Scope of Phase 1

Phase 1 produces a BOAMP-only output table: one row per eligible *APPEL_OFFRE* with a binary column **event**. In this report, **event** = 1 means that an identifiable reappearance of a similar procurement need was found by the selected linking rule, while **event** = 0 means that no such link was observed under that rule. After the 2026-07-08 method-comparison experiment, the current Phase-2 handoff is `boamp_phase2_survival_method_m0_balanced.csv`. The older `boamp_phase2_survival.csv` file is retained as the historical pre-calibration 665-event baseline.

2 Input Data and Column Dictionary

2.1 Source and Extraction

The dataset was extracted from the BOAMP API and covers notices published from **2015-03-02 to 2024-12-29**. The raw file after initial cleaning is `data/processed/boamp_full_clean.csv`.

Table 4: Dataset overview

Notice type	Count	Share
APPEL_OFFRE (calls for tender)	1,933	60.8%
ATTRIBUTION (award notices)	1,081	34.0%
RECTIFICATIF (corrections)	150	4.7%
PRE-INFORMATION	7	0.2%
MODIFICATION	5	0.2%
INTENTION_CONCLURE	4	0.1%
PERIODIQUE	1	0.0%
Total	3,181	100%

The renewal linking algorithm operates exclusively on the 1,933 *APPEL_OFFRE* notices. *ATTRIBUTION* notices are used only as a secondary source to extract more precise contract start dates (via the `annonce_lie` back-reference field).

2.2 Column Dictionary — Raw Dataset (44 columns)

Table 5 describes all 44 columns in `boamp_full_clean.csv`. Columns are grouped by origin: raw BOAMP fields and fields added by the cleaning pipeline.

Table 5: Full column dictionary — `boamp_full_clean.csv`

Column	Type	Coverage	Description
— <i>Identifiers and dates</i> —			
<code>idweb</code>	string	100%	Unique BOAMP notice identifier (e.g. 22-111359). Primary key.
<code>dateparution</code>	date	100%	Official publication date of the notice. Used as contract start proxy when no award date exists.
<code>datefindiffusion</code>	date	100%	End-of-diffusion date (administrative). Not used in modelling.
<code>datelimitereponse</code>	datetime	42.8%	Deadline for tender submission. Only partially filled.
<code>date_attribution</code>	date	0%*	Award date parsed from the notice body (0% on AO records; populated only on <i>ATTRIBUTION</i> notices).
<code>date_publication_anterieure</code>	date	0.4%	Reference to a prior publication. Sparse; not used.
— <i>Buyer information</i> —			
<code>nomacheteur</code>	string	100%	Raw buyer name as published. Inconsistent casing and accents.
<code>code.departement</code>	string	100%	French department code (e.g. 44 for Loire-Atlantique).
<code>buyer_siret</code>	string	9.1%	SIRET number (14-digit unique company ID). Low coverage.
<code>buyer_cp</code>	string	96.5%	Postal code of the buyer.
<code>buyer_ville</code>	string	99.6%	City name of the buyer.

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Table 5 (continued)

Column	Type	Coverage	Description
buyer_key [†]	string	100%	Constructed key: SIRET:XXXXXXX when SIRET is available, else NAME:normalised_name. Used as the grouping key for renewal matching.
— Contract classification —			
famille	string	100%	BOAMP family code (JOUÉ = EU Journal, or other).
nature	string	100%	Notice type (see Table 1 above).
nature_libelle	string	100%	Human-readable notice type label.
type_procedure	string	98.5%	Procurement procedure: OUVERT (62.4%), PROCEDURE_ADAPTE (24.3%), NEGOCIE (8.9%), RESTREINT (1.9%), other.
type_marche	string	91.6%	Contract type: SERVICES (53.6%), FOURNITURES (32.0%), TRAVAUX (5.4%), mixed.
type_avis	string	95.1%	Notice subtype codes (e.g. 5111). 100% on AO notices.
etat	string	100%	Publication state (INITIAL 97.6%, ANNULE, RECTIFIE).
contractfolderid	string	9.8%	Internal contract folder ID. Very sparse; not used.
— Contract description —			
objet	string	100%	Free-text contract description. Mean length: 105 characters (range: 6–201). The primary text signal for semantic matching.
descripteur_libelle	string	100%	Thematic descriptor in natural language (e.g. “Informatique (prestations de services)”). 549 unique values.
titulaire	string	0%	Winning contractor (populated only on ATTRIBUTION, not AO).
annonce_lie	string	2.9%	Back-reference to an earlier notice. On ATTRIBUTION records: 82.9% filled. Used as indirect external calibration signal (back-reference on ATTRIBUTION notices).
— CPV codes —			
cpv_principal	float	98.1%	Raw CPV code (8-digit integer) as published.
cpv_all	string	100%	Pipe-delimited list of all CPV codes for the notice.
cpv_clean [†]	float	98.1%	Cleaned primary CPV code. Generic codes (ending in 000000) flagged separately.
cpv_div2 [†]	float	98.1%	First 2 digits of CPV (division-level, e.g. 72 = IT services). Range: 15–98.
cpv_class4 [†]	float	98.1%	First 4 digits of CPV (class-level granularity).
category_id [†]	string	100%	Internal category code (CAT01–CAT10, plus CAT_UNKNOWN) derived from CPV.
category_label [†]	string	100%	Human-readable category (see Table 6).
— Contract amount —			
amount_eur	float	43.0%	Raw amount in euros as parsed from the notice.
amount_raw	float	43.0%	Amount before cleaning transformations.
amount_clean [†]	float	41.1%	Cleaned amount: zero-values, suspiciously small values, and ceiling-flagged amounts removed. Range: €2,000 – €9.6M; median €350,000.
flag_amount_zero [†]	int	100%	1 if amount_eur = 0 (reported but missing).
flag_amount_tiny [†]	int	100%	1 if amount < €1,000 (likely misreported).
flag_amount_ceiling [†]	int	100%	1 if amount ≥ €10,000,000 (likely a framework ceiling, not actual spend).
— Contract duration —			
duration_months	float	47.7%	Duration in months as parsed from the notice.
duration_raw_boamp	float	47.7%	Raw parsed value before cleaning.

Continued on next page

Table 5 (continued)

Column	Type	Coverage	Description
<code>duration_source_field</code>	string	47.7%	Source field from which duration was extracted (<code>DUREE_MOIS</code> , <code>DUREE_JOURS</code> , etc.).
<code>flag_duration_suspect</code> [†]	int	100%	1 if duration > 120 months (10 years) or = 0.
<code>duration_clean</code> [†]	float	47.3%	Cleaned duration: suspect values set to missing. Range: 1–120 months; median 24 months (among non-imputed).
— <i>Other</i> —			
<code>url_avis</code>	string	100%	Direct URL to the notice on boamp.fr.
<code>donnees_format</code>	string	100%	Data format flag (<code>legacy</code> 87.3%, <code>json</code> 12.7%).
[†] Column added or derived by the cleaning pipeline. Not present in the original BOAMP API response.			
* <code>date_attribution</code> is 0% for AO notices; it is populated on <code>ATTRIBUTION</code> notices and used indirectly via <code>annonce_lie</code> .			

2.3 Contract Category Distribution

The 10 internal categories (CAT01–CAT10, plus an Unknown bucket for missing CPV) are derived from the CPV classification and cover exclusively the ICT/digital services scope of the Gigalis portfolio.

Table 6: Category distribution among APPEL-OFFRE notices

Category ID	Label	Count	Share
CAT02	IT Services & Consulting	583	30.2%
CAT01	Software & Applications	373	19.3%
CAT04	Telecom & Networks	337	17.4%
CAT03	Cybersecurity	211	10.9%
—	Unknown (CPV missing)	176	9.1%
CAT07	Digital Workplace & Collab.	89	4.6%
CAT08	Data & AI	54	2.8%
CAT06	IT Hardware & Equipment	45	2.3%
CAT10	IT Maintenance & Support	31	1.6%
CAT05	Cloud & Infrastructure	25	1.3%
CAT09	GIS & Mapping	9	0.5%
Total AO		1,933	100%

2.4 Buyer Landscape

Before SIREN enrichment, the dominance of name-based buyer keys (87.2%) was the primary structural risk: variant spellings of the same buyer (e.g. “Nantes” vs. “Nantes Métropole” vs. “NM”) fragment the buyer group, creating structural misses that the algorithm cannot recover.

Table 7: Buyer concentration among APPEL_OFFRE notices

Metric	Value
Total unique buyers (<code>buyer_key</code>)	475
Buyers identified by SIRET	61 (12.8%)
Buyers identified by normalised name only	414 (87.2%)
Buyers with ≥ 2 AO notices	225 (47.4%)
Buyers with ≥ 5 AO notices	78 (16.4%)
<i>Top 5 buyers by number of AO notices</i>	
Nantes Métropole	128
Des Pays de la Loire	109
De Nantes	61
Saint-Nazaire	50
Angers Loire Métropole	46

2.5 Data Source Comparison: BOAMP vs. DECP

Table 8 provides a field-level comparison of both sources within scope (PdL digital contracts, 2015–2024). This informed the primary/enrichment architecture adopted for the rest of the analysis.

Table 8: Field availability and completion by source ($n_{\text{BOAMP}} = 3,181$; $n_{\text{DECP}} = 3,039$)

Field	BOAMP		DECP		Recommended source
	Avail.?	Fill%	Avail.?	Fill%	
Buyer SIRET	Partial	9.1	Yes	100.0	DECP
Buyer name	Yes	100.0	Yes	100.0	Both
Contract object (text)	Yes	100.0	Yes	99.9	BOAMP
CPV code	Yes	98.1	Yes	100.0	Both
Amount (EUR)	Partial	43.0	Yes	95.6	DECP
Contract duration	Partial	47.7	Yes	99.4	DECP
Notification date	No	—	Yes	100.0	DECP
Publication date	Yes	100.0	Yes	98.0	BOAMP
Award date	Partial	31.8	No	—	BOAMP
Winner identity	Partial	27.6	Yes	95.5	Both
Procedure type	Yes	92.9	Yes	85.4	Both
Offers received	No	—	Partial	26.4	DECP
Notice type (AAPC/award)	Yes	100.0	No	—	BOAMP
Linked notices	Partial	34.9	No	—	BOAMP

Architecture decision. BOAMP was retained as the primary corpus: it is the only source covering the full 2015–2024 period, it distinguishes contract notices from award notices (the event structure survival analysis requires), and provides the richest free-text objects for Phase-2 NLP classification. Although DECP was explored as a possible enrichment source, the current official handoff remains BOAMP-only because BOAMP and DECP do not share a robust direct contract-level linking key. DECP therefore remains a later enrichment path rather than part of the present modeling dataset.

2.6 Duration Field: Scientific Reliability Analysis

Scientific Question

Is the declared duration field (`duration_clean`) reliable as a survival target?

The answer is **no**. The declared duration is preserved as a covariate and as a temporal search prior for the linking algorithm, but it is not the survival target. The true survival target — the actual time elapsed between a contract start and its renewal — is the linked gap $g(i, j^*) = t_{j^*} - t_i$, stored in `observed_duration_months` in the output file.

Evidence

Of the 1,933 *APPEL_OFFRE* notices, 1,480 (76.6%) have a valid, non-suspect `duration_clean` value. Table 9 shows the distribution of these values.

Table 9: Distribution of declared duration values (non-missing AO, $n = 1,480$)

Duration (months)	Count	Share	Cumulative
48	462	31.2%	31.2%
12	316	21.4%	52.6%
24	151	10.2%	62.8%
36	99	6.7%	69.5%
<i>Subtotal: exactly 12/24/36/48 months</i>	<i>1,028</i>	<i>69.5%</i>	
60	70	4.7%	74.2%
4	67	4.5%	78.7%
6	52	3.5%	82.2%
72	20	1.4%	83.6%
(other values)	243	16.4%	100.0%
Total	1,480	100%	

Mean: 30.8 months; Median: 24 months; Std: 21.2 months.

The diagnostic signal is the extreme clustering at round multiples of 12. Real contract durations driven by operational timelines would produce a continuous distribution with noise. Instead, nearly 70% of all values fall on exactly four integers — all multiples of 12 months — and the next clear spikes are the administrative 5-year and 6-year ceilings (60 and 72 months), which again reflect legal maxima rather than empirically measured durations.

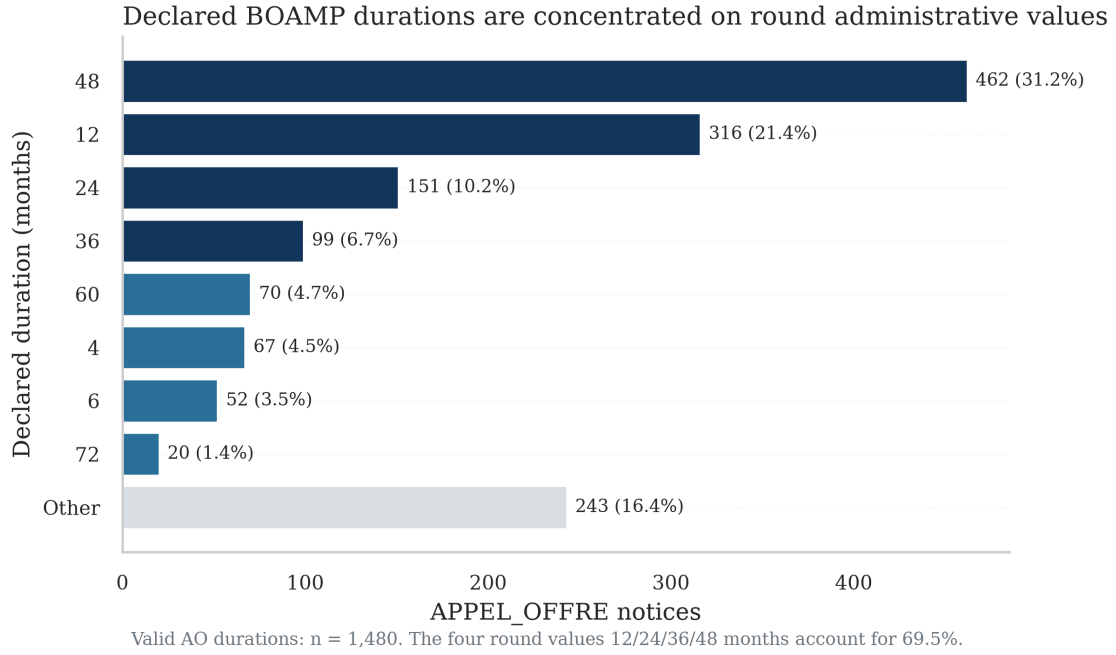


Figure 7: Distribution of declared duration values for the 1,480 *APPEL_OFFRE* notices with a valid `duration_clean`. Dark bars mark the four exact round values (12, 24, 36, 48 months) that together account for 69.5% of all valid durations. Light-blue bars are other specific values; grey is all remaining values combined. The concentration at round multiples of 12 reflects the administrative maximum declaration required by French procurement law, not empirically measured contract lifetimes.

Explanation

French public procurement law (Code de la commande publique) requires the buyer to declare the **administrative maximum duration** of the contract at the time of publication. This is defined as the base period plus all possible renewal tranches stacked end-to-end. A buyer writing a 12-month base contract with three possible 12-month renewal options declares `duration_months = 48`. The 48 months is a legal ceiling approved at signature, encoding the outer boundary of the contract, not a forecast of when the buyer intends to re-enter the market.

Using this field as the survival target would mean training the model on administrative ceilings rather than on observed behaviour. An AO that declares 48 months but renews after 12 months (at the end of the first base period) would be incorrectly labelled as a 48-month contract. The error is not random noise: it is systematically biased upward, because buyers default to the maximum configured option when completing the publication form.

Resolution: What the Field Is Used For

The declared duration d_i is retained for three specific purposes:

1. **Search prior in the linking algorithm.** The estimated end date $\hat{e}_i = t_i + d_i$ defines

the centre of the temporal candidate window (Section 2.6 and Step 2 of the algorithm in Section 5). The declared duration tells the algorithm *where to look* for a renewal notice, not *what the answer is*.

2. **Covariate in survival models.** The administrative duration choice encodes the contractual structure the buyer selected (single-year vs. multi-year framework). This is predictive of renewal behaviour even when the raw value does not equal the actual time-to-renewal.
3. **Right-censoring indicator.** Contracts for which $\hat{e}_i = t_i + d_i > T_{\max}$ (December 2024) are excluded from the eligible denominator: their renewal window is not observable in the study period, regardless of algorithm quality.

The **actual survival target** is the observed gap:

$$T_i^{\text{obs}} = g(i, j^*) = t_{j^*} - t_i \quad (\text{months}) \quad (1)$$

stored in `observed_duration_months` in `boamp_renewal_links.csv`. This quantity is available only for the 665 linked contracts (`event = 1`) and is the correct input duration for the Cox and Kaplan–Meier models in Phase 2. For right-censored contracts (`event = 0`), the survival model requires a censoring time, which is typically taken as the study end date minus the contract start date.

3 Dataset Construction and Missing-Value Control

This section explains the dataset in operational terms: what each row means, why each intermediate table exists, how the modeling population is constructed, and how incomplete fields are controlled. The objective is not only to document the pipeline, but also to make the dataset explainable during a technical discussion.

3.1 What Each Dataset Represents

The project does not start from a ready-made table of contract renewals. BOAMP publishes **notices**, not longitudinal contract histories. A BOAMP row is one publication event: a call for tenders, an award notice, a correction, or another administrative notice. The analysis therefore transforms a notice-level source into a contract-level survival dataset.

1. **Raw notice corpus.** The API extraction and flattening step produces `boamp_full_flat.csv` with 3,181 BOAMP notices and 30 raw/flattened columns.
2. **Cleaned analytical corpus.** The cleaning step produces `boamp_full_clean.csv` with 3,181 notices and 44 columns. This file keeps the raw variables and adds controlled fields such as `buyer_key`, `cpv_clean`, `amount_clean`, `duration_clean`, and taxonomy labels.

3. **Source contract population.** Only the 1,933 APPEL_OFFRE notices are used as possible source contracts, because they represent tenders that can later be re-tendered. Award, correction, and modification notices are informative context but are not survival units.
4. **Eligible modeling population.** A source contract enters the final linking denominator only if its expected renewal window is observable before the study end date, 2024-12-31 (with the $W = 6$ -month search window, this requires an estimated end date on or before 2024-06-30). This gives 1,210 eligible contracts; the other 723 source notices are too recent to be evaluated fairly.
5. **Survival handoff.** The final file, `boamp_phase2_survival.csv`, has one row per eligible contract, with `event`, `observed_duration_months`, renewal diagnostics, and covariates used by Phase 2.

This distinction is important. The 3,181 rows describe the available BOAMP evidence; the 1,933 APPEL_OFFRE rows describe possible source contracts; the 1,210-row handoff is the population on which the official renewal-linking and survival results are measured.

3.2 How Missing Values Are Handled

The guiding rule is **flag first, preserve raw values, impute only when a downstream algorithm truly needs a value**. Missingness is treated as information about data quality, not as something to hide.

Table 10: Main missing-value and quality-control decisions

Field	Observed quality	Handling rule	Reason
Buyer identity	BOAMP SIRET is filled on 9.8% of AO notices; only 5.8% of AO rows become SIRET-anchored keys after validation.	Use SIRET:... when a valid 14-digit SIRET exists; otherwise use NAME:... from a normalised buyer name.	A weak name key is still necessary because renewal matching requires buyer grouping. The limitation is made explicit rather than discarded.
Contract amount	Only 21.4% of AO notices have a valid amount_clean ; the full BOAMP corpus has 41.1% coverage.	Keep amount_eur and amount_raw ; set amount_clean to missing when the value is zero, below €1,000, or at least €10M.	Zero and tiny amounts are likely encoding errors; very large values are often framework ceilings rather than actual spend. Amount is therefore not used as a core matching signal.
Contract duration	76.6% of AO notices have a clean declared duration; 5 AO durations are flagged as suspect.	Keep the raw duration; set suspect values outside 1–120 months to missing. For linking and survival-window construction only, missing duration is imputed to 48 months and dur_was_imputed records this choice.	The renewal window cannot be constructed without a duration. The 48-month value is a transparent default, not a claim that the true duration is known.
CPV code	96.8% of AO notices have a cleaned CPV code.	Derive the full CPV hierarchy (cpv_full18 , cpv_div2 , cpv_group3 , cpv_class4 , cpv_category5) and category labels when CPV exists. When CPV is missing in a candidate pair, it is excluded from the composite score and the remaining component weights are renormalized.	A missing CPV should not automatically erase a plausible renewal when text, buyer, and timing agree, but it must not inject an arbitrary neutral signal either.
Start date	AO notices do not directly carry a complete award-date field.	Use a linked ATtribution notice date when available; otherwise use dateparution as the start-date proxy.	Publication date is imperfect but observable for every notice; attribution back-references improve precision where they exist.
Renewal outcome	BOAMP has no explicit renewal label.	Build event by linking a later notice from the same buyer with compatible text, CPV, and timing. If no accepted link is found before study end, set event=0 and treat the row as right-censored.	event=0 is not proof of no renewal; it means no renewal was observed under the algorithm and available time window.

3.3 Practical Examples From Current Rows

Table 11 gives concrete examples from the current BOAMP handoff. These examples are useful because they show how the same rule is applied in different situations: a row can have missing amount, missing CPV, or an imputed duration and still be usable, provided the uncertainty is carried forward explicitly.

Table 11: Examples of raw-to-modeling transformations

Example	Raw evidence	Pipeline decision	Interpretation
Missing amount, still usable	15-31464: SDIS de la Mayenne, object about small fire and rescue equipment. Amount and CPV are missing; declared duration is 36 months.	<code>amount_clean = NaN</code> ; <code>category_label = Unknown</code> ; <code>duration_clean = 36</code> . The row remains eligible because buyer, text, and timing are still available.	Missing amount is a data-quality limitation, not a reason to delete the contract. Amount is not a core matching signal.
Imputed duration, explicitly flagged	15-33241: CHU de Nantes, acquisition and maintenance of an “appel malade” system. Raw duration is missing.	Duration is set to 48 months only for the renewal-window search, and <code>dur_was_imputed = True</code> . The row links to 20-19986 after 52.96 months.	The model can use the row, but the imputation flag tells us that the temporal prior is weaker than for an observed duration.
Censored contract, not bad data	15-30947: SMICTOM de la Vallee de l’Authion, software and equipment for waste collection. Amount is valid (€150,000) and declared duration is 12 months.	No accepted renewal is found. The handoff stores <code>event = 0</code> , <code>renewal_duration_months = NaN</code> , <code>censoring_duration_months = 118.00</code> , and <code>observed_duration_months = 118.00</code> .	The row says: “we observed this contract for 118 months without an accepted renewal link.” It is valid survival information.
SIRET fallback to name key	The same three examples have no valid BOAMP buyer SIRET.	The pipeline uses <code>NAME:de la mayenne</code> , <code>NAME:de nantes</code> , or <code>NAME:smictom de la vallee de l authion</code> .	This makes linking possible, but it also explains a risk: if the same buyer appears under another spelling, the buyer group may split.

Worked link example. For 15-31464, the source notice is published on 2015-03-02 with a 36-month declared duration. The algorithm searches for later notices from the same buyer around the expected renewal period. It selects 17-176560 as the best candidate with `text_similarity` = 0.6922, `cpv_match_score` excluded (source CPV missing, `cpv_used_in_score` = False), `temporal_score` = 0.7951, and `composite_score` = 0.7812. The final survival row is therefore `event` = 1 with `renewal_duration_months` = 33.54. This is a good example of why the method uses several signals: the CPV field is incomplete, but the buyer, text, and timing evidence are still strong enough to create a high-confidence link.

Worked censoring example. For 15-30947, the contract starts on 2015-03-02 and has a clean 12-month declared duration and a valid amount. The algorithm does not find an accepted renewal candidate before 2024-12-31. The row is not removed; it becomes `event` = 0 with a censoring duration of 118.00 months. In survival language, the only statement is: “the renewal time is longer than the observed follow-up, or the renewal was not detected.” This is why the row remains in Kaplan–Meier and Cox modeling.

3.4 Why Right-Censoring Is Not Missing Data

The final dataset contains 665 linked renewals and 545 unlinked eligible contracts. The unlinked rows remain in the survival dataset because they still carry valid information: they survived without an observed renewal until the last date we could observe them. In survival-analysis notation, those rows have `event` = 0 and a censoring duration. Removing them would bias the model toward contracts that renewed quickly and would make long-running or late-window contracts disappear from the risk set.

The interpretation is therefore:

- `event` = 1: the algorithm found a plausible later renewal notice; `renewal_duration_months` is observed.
- `event` = 0: no accepted renewal was observed before study end; `censoring_duration_months` is observed.
- `observed_duration_months`: always filled, using the renewal gap for events and the time-to-study-end for censored rows.

This is why the 1,210-row handoff is suitable for Kaplan–Meier and Cox models: every row contributes time-at-risk, even when the renewal event itself was not observed.

3.5 Practical Explanation of the Pipeline

In plain terms, the dataset is built as follows. First, the project collects all relevant digital-procurement BOAMP notices in Pays de la Loire from 2015 to 2024. Second, it cleans the

unstable fields without overwriting the originals: buyer names become canonical keys, CPV codes become comparable groups, amounts and durations are flagged when implausible, and each contract receives a technology category. Third, it keeps only call-for-tender notices as contracts that can later be renewed. Fourth, it searches forward in time for later notices from the same buyer that look like the same service being re-tendered. Finally, it writes a survival table where each contract has a start date, a duration, an event indicator, and enough diagnostics to judge whether the inferred renewal link is strong or weak.

The most important limitation to remember is that the target variable is constructed, not directly observed. The strength of the project is that this construction is explicit: each assumption is visible in the output columns and can be audited, adjusted, or tested through sensitivity analysis.

4 Data Quality Issues Diagnosed

Nine data quality issues were identified through systematic EDA. Each directly affects the accuracy or recall of the renewal linking algorithm.

Table 12: Data quality issues and their impact on the linking algorithm

#	Issue	Observation	Impact on linking
1	SIRET coverage low	Only 5.8% of <i>APPEL_OFFRE</i> notices have a SIRET-anchored buyer_key . The SIRET <i>field</i> is present on 9.8% of AO (9.1% across all 3,181 notices), but $\sim 46\%$ of those values fail validation and fall back to the name key; 12.8% of the 475 unique buyer keys are SIRET-anchored. The rest use normalised name as buyer key	Variant spellings split buyer groups $\Rightarrow$ structural misses
2	Buyer name variants	Same buyer appears under multiple spellings after normalisation	False negative links where buyers are split
3	CPV missing or generic	3.2% of AO have no CPV; generic catch-all codes are capped at the division level	Missing CPV is excluded from the composite and the remaining weights renormalized
4	Contract amount sparse	Only 21.4% of AO report a valid, non-flagged amount	Amount cannot be used as a matching signal
5	Duration missing	23.4% of AO have no duration; imputed to 48 months	Wrong imputed duration shifts the temporal window
6	Duration suspect values	Some raw durations > 120 months or $= 0$	Removed; fall into imputation bucket
7	Contract start date imprecise	date_attribution is 0% on AO records; publication date used as proxy for 59.3%	Up to several weeks of noise in temporal scoring
8	Short or generic <i>objet</i>	24 notices have objet < 20 characters	Semantic embeddings less discriminative for short texts
9	annonce_lie interpretation	Present only on ATtribution notices (82.9%); refers to same-contract back-link, not a renewal	Used as calibration signal, not as a direct renewal label

5 Preprocessing Pipeline

5.1 Buyer Key Construction and SIREN Enrichment

$$k_i^{(0)} = \begin{cases} \text{SIRET}:r_i & \text{if } r_i \in \{0, \dots, 9\}^{14} \\ \text{NAME}:\phi(n_i) & \text{otherwise} \end{cases} \quad (2)$$

For notice i , n_i is the raw BOAMP buyer name (`nomacheteur`), r_i is the raw BOAMP buyer SIRET (`buyer_siret`), and $k_i^{(0)}$ is the initial buyer key used before SIREN enrichment. SIRET is the 14-digit establishment identifier; SIREN is the 9-digit legal-entity identifier. BOAMP SIRET coverage is sparse, so the initial rule is deliberately SIRET-first and name-fallback: use a precise establishment identifier when it is valid, otherwise keep the row by grouping it under a normalised name. The normalisation function $\phi(\cdot)$ lowercases, strips accents, removes punctuation, removes legal or administrative suffixes when matched by the local rule set, and collapses whitespace.

Unique-buyer table. The enrichment pipeline does not query the API once per notice. It first constructs one row per distinct `nomacheteur`. Let $\mathcal{U}$ be the set of unique raw buyer names and let $I(u) = \{i : n_i = u\}$ be the notices carrying buyer name u . For each u , the pipeline stores:

$$x_u = (u, \phi(u), \text{mode}\{k_i^{(0)} : i \in I(u)\}, \delta_u, v_u, p_u, |I(u)|, |\{i \in I(u) : \text{AO}\}|), \quad (3)$$

where δ_u is the modal department code, v_u the modal city, and p_u the modal postal code. Existing BOAMP SIRET values are preserved as audit metadata but do not by themselves create a new SIREN key. This produces 525 unique buyer rows from the 3,181-notice corpus.

API candidate retrieval. For each $u \in \mathcal{U}$, the code detects a public-entity type a_u from name patterns such as `COMMUNE`, `SDIS`, `CHU`, `REGION`, `EPCI`, or `DEFAULT`. This determines both a short query string q_u and, when known, a French legal-nature filter η_u (`nature_juridique`). The API search is attempted in a controlled sequence:

1. query q_u with department δ_u and legal-nature filter η_u (except `SDIS`, where the legal-nature filter is skipped because it returned no results in this API);
2. if no candidates are returned, query q_u with the department only;
3. if still empty, query q_u with the legal-nature filter only;
4. if still empty and the raw normalised name differs from q_u , query the raw normalised name with the department only.

The API returns at most 10 active-entity candidates per buyer. Each candidate c_{ur} has rank r , SIREN σ_{ur} , establishment SIRET τ_{ur} , official name m_{ur} , NAF activity code, legal-nature code, city, and postal code. All returned candidates are written to `boamp_buyer_siren_candidates.csv`; in the current run this table has 1,514 real API candidate rows for 401 of the 525 buyer names. Results are cached by normalised buyer name and department so the pipeline is reproducible without repeatedly calling the API.

Candidate scoring. The candidate score is a name-resolution score, not a procurement-renewal score. Let $\psi(\cdot)$ denote the SIREN scoring normalisation: accents, case, punctuation, legal suffixes, and common French stopwords are removed. Let h_u be the comparison string: for most buyers $h_u = \psi(u)$, but special forms are rewritten to match the SIRENE naming convention, for example `Ville de X` becomes `commune X` and `SDIS` is compared to `sce departemental incendie secours sdis`. For candidate c_{ur} ,

$$R(u, r) = \text{token_sort_ratio}(h_u, \psi(m_{ur})), \quad (4)$$

$$O(u, r) = \begin{cases} \frac{|\text{Tok}(h_u) \cap \text{Tok}(\psi(m_{ur}))|}{|\text{Tok}(h_u)|} & \text{if } |\text{Tok}(h_u)| > 0, \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

$$E(u, r) = \min\{100, R(u, r) + 8O(u, r)\}. \quad (6)$$

$R(u, r)$ is RapidFuzz token-sort similarity on $[0, 100]$, $O(u, r)$ is the fraction of query tokens recovered in the official candidate name, and $E(u, r)$ is the stored `enrichment_score`. The API rank is preserved: the proposed match is the first returned API candidate, and the score and margin are used as quality filters. If n_u candidates are returned, define

$$e_u^{(1)} = E(u, 1), \quad e_u^{(2)} = \begin{cases} E(u, 2) & \text{if } n_u \geq 2, \\ \text{NaN} & \text{if } n_u = 1, \end{cases} \quad \Delta_u = e_u^{(1)} - e_u^{(2)}. \quad (7)$$

Confidence classification and key upgrade. Confidence is assigned at the unique-buyer level:

$$\text{conf}(u) = \begin{cases} \text{HIGH} & e_u^{(1)} \geq 85, \\ \text{HIGH} & e_u^{(1)} \geq 75 \wedge n_u = 1, \\ \text{HIGH} & e_u^{(1)} \geq 75 \wedge \Delta_u \geq 10, \\ \text{MEDIUM} & e_u^{(1)} \geq 70 \text{ and not HIGH,} \\ \text{LOW} & e_u^{(1)} \geq 55 \text{ and not MEDIUM,} \\ \text{NO_MATCH} & \text{otherwise.} \end{cases} \quad (8)$$

Only HIGH-confidence matches with a valid SIREN change the grouping key:

$$k_u^{(1)} = \begin{cases} \text{SIREN}:\sigma_{u1} & \text{if } \text{conf}(u) = \text{HIGH} \text{ and } \sigma_{u1} \text{ exists,} \\ k_u^{(0)} & \text{otherwise.} \end{cases} \quad (9)$$

Thus MEDIUM and LOW matches are diagnostic only; they are not allowed to merge buyer groups. Buyers already keyed by BOAMP SIRET are also left auditable as SIRET-origin cases. The enriched key $k_u^{(1)}$ is left-joined back to all BOAMP notices by exact `nomacheteur`, preserving row count. In the current run, 275 of 525 unique buyer names are HIGH confidence (52.4%), 21 MEDIUM, 32 LOW, and 197 NO_MATCH. These HIGH matches create 275 `SIREN`: upgrades and merge 23 previously fragmented buyer groups. In the final 1,210-row survival handoff, 724 rows use a `SIREN`: key, 484 retain a `NAME`: key, and 2 retain a `SIRET`: key.

Use in renewal linking. After enrichment, the active buyer identifier in the renewal-linking algorithm is

$$b_i = k_{n_i}^{(1)}. \quad (10)$$

The matching logic is otherwise unchanged: SIREN enrichment changes the blocking key, not the text, CPV, temporal, or composite-score formulas. This is important for interpretation. A SIREN merge can add renewal candidates because two spelling variants now share the same legal-entity key, but a HIGH-confidence SIREN match is still automated entity resolution, not manual ground truth.

5.2 CPV Normalisation and Taxonomy Tagging

The taxonomy step converts heterogeneous CPV and text descriptions into a single ICT segment label used in descriptive tables, bias checks, and survival covariates. Let x_i be the raw primary CPV value and o_i the BOAMP *objet*. The CPV cleaning function $\kappa(\cdot)$ is:

$$c_i = \kappa(x_i), \quad (11)$$

where κ drops the CPV check digit after a dash, removes numeric-format artefacts such as `.0`, keeps the first 8 digits, restores a lost leading zero for 7-digit numeric storage, and returns missing when no usable digits remain. From the cleaned 8-digit code c_i , the pipeline derives

$$c_i^{(2)} = c_i[1:2], \quad c_i^{(3)} = c_i[1:3], \quad c_i^{(4)} = c_i[1:4], \quad c_i^{(5)} = c_i[1:5], \quad c_i^{(8)} = c_i, \quad (12)$$

stored respectively as `cpv_div2`, `cpv_group3`, `cpv_class4`, `cpv_category5`, and `cpv_full8`.

A code ending in 000000 is flagged as generic because it identifies only a broad division, not a specific service.

The taxonomy file `data/processed/taxonomy.csv` defines 10 ICT categories. For category q , let P_q be its set of CPV prefixes and K_q its keyword regular expression. The assignment is CPV-first:

$$L(i, q) = \max_{p \in P_q} \{|p| : c_i \text{ starts with } p\}, \quad L(i, q) = 0 \text{ if no prefix matches.} \quad (13)$$

$$q_i^{\text{cpv}} = \arg \max_q L(i, q) \quad \text{if } \max_q L(i, q) > 0. \quad (14)$$

The longest matching prefix wins. If two categories would tie at the same prefix length, the first category order in `taxonomy.csv` is kept, because the implementation updates the best match only when a strictly longer prefix is found. If no CPV prefix matches, the pipeline falls back to the contract description:

$$q_i^{\text{text}} = \min_{\text{taxonomy order}} \{q : K_q(o_i) = 1\}. \quad (15)$$

The final category is therefore:

$$q_i = \begin{cases} q_i^{\text{cpv}} & \text{if } \max_q L(i, q) > 0, \\ q_i^{\text{text}} & \text{if no CPV prefix matches and a keyword matches,} \\ \text{CAT_UNKNOWN} & \text{otherwise.} \end{cases} \quad (16)$$

This rule is intentionally conservative: structured CPV evidence has priority, keyword matching only rescues rows without a usable CPV-prefix match, and no row is forced into a digital category when both signals are absent. In `boamp_full_clean.csv`, CPV is cleaned for 3,119 of 3,181 notices (98.1%). The keyword fallback assigns a non-Unknown category to 42 notices whose cleaned CPV is missing; the remaining unresolved notices stay in the Unknown bucket.

5.3 Contract Start Date Refinement

Award date from linked ATTRIBUTION notices is used when available, falling back to the publication date:

$$t_i = \begin{cases} \text{award_date}(j) & \text{if } \exists j \in \text{ATTR} : \text{annonce_lie}(j) = i \\ \text{dateparution}_i & \text{otherwise} \end{cases} \quad (17)$$

In practice, 40.7% of eligible AO obtain a more precise start date through this refinement.

5.4 Duration Imputation

$$d_i = \begin{cases} \text{duration_clean}_i & \text{if not missing and not suspect} \\ d_0 = 48 \text{ months} & \text{otherwise (23.4\% of AO)} \end{cases} \quad (18)$$

5.5 Estimated End Date

$$\hat{e}_i = t_i + d_i \quad (19)$$

5.6 Text Normalisation

The *objet* field is normalised for TF-IDF (used in calibration) and passed raw to Sentence-Transformers:

1. Lowercase
2. Strip Unicode accents (NFD normalisation)
3. Keep only characters matching [a-z0-9]
4. Collapse multiple spaces

5.7 Sentence-Transformer Embeddings

Each normalised *objet* $\mathbf{o}_i$ is encoded into a 384-dimensional unit-norm vector:

$$\mathbf{v}_i = \text{ST}(\mathbf{o}_i) \in \mathbb{R}^{384}, \quad \|\mathbf{v}_i\|_2 = 1 \quad (20)$$

Model used: `paraphrase-multilingual-MiniLM-L12-v2` (multilingual, 384 dimensions, ~120 MB).

5.8 Cleaning Statistics Summary

Tables 13–15 quantify the impact of the Week-2 cleaning rules applied to both sources.

Table 13: Amount cleaning flags and fill rates by source

Source	Zero (= 0)	Tiny (<€1 k)	Ceiling ($\geq$ €10 M)	<code>amount_clean</code> fill
BOAMP ($n = 3,181$)	1	9	53	41.1%
DECP ($n = 3,039$)	31	20	23	93.1%

Flagged values are set to `NaN` in `amount_clean` but the raw field is preserved. The low BOAMP fill rate reflects the structural gap documented in Table 8: amount reporting was mandatory only for eForms notices (2024+). No amount imputation is applied; median-by-segment imputation is deferred to Phase 3 modelling.

Table 14: Duration cleaning and fill rates by source

Source	Suspect flags (outside [1, 120] months)	<code>duration_clean</code> fill
BOAMP ($n = 3,181$)	12	47.3%
DECP ($n = 3,039$)	6	99.2%

Table 15: Buyer key normalization results

Source	Raw unique names	Canonical keys	SIRET-anchored
BOAMP	525	502	156 rows (4.9%)
DECP	278	294	3,039 rows (100.0%)

The cross-source buyer bridge is saved to `buyer_bridge.csv`. Name normalisation applies lowercase, accent stripping, punctuation removal, and legal-form suffix removal (e.g. *Commune de, SARL*).

6 Renewal Linking Algorithm

6.1 Notation

Let $\mathcal{A}$ denote the set of all *APPEL_OFFRE* notices. Each notice $i \in \mathcal{A}$ is represented by:

- t_i : contract start date, defined as the award date when available, otherwise the BOAMP publication date;
- d_i : contract duration in months, after cleaning and imputation;
- $\hat{e}_i = t_i + d_i$: estimated contract end date;
- CPV_i : main CPV classification code;
- $\mathbf{o}_i$: contract description text (*objet*);
- b_i : buyer identifier, using SIRET when available and normalised buyer name otherwise.

The goal is to construct, for each eligible source notice i , an algorithmic binary outcome Y_i indicating whether a plausible renewal candidate can be found among later BOAMP calls-for-tender.

6.2 Step 1 — Eligibility Filter

A notice is **eligible** to be a renewal source if its full renewal search window (estimated end date $\pm W$ months, $W = 6$) closes within the study period $[T_{\min}, T_{\max}]$:

$$i \in \mathcal{A}_{\text{elig}} \iff \hat{e}_i + W \leq T_{\max} \quad (T_{\min} = 2015-01-01, T_{\max} = 2024-12-31), \quad (21)$$

i.e. the estimated end date must fall on or before 2024-06-30.

Table 16: Eligibility breakdown

Population	Count
Total APPEL_OFFRE	1,933
Not eligible: renewal window closes after 2024-12-31	723
Eligible ($\mathcal{A}_{\text{elig}}$)	1,210

Contracts with $\hat{e}_i > T_{\max}$ are excluded from the Phase 1 linking denominator because the renewal opportunity is not sufficiently observable in the 2015–2024 dataset.

6.3 Step 2 — Candidate Pool

For each eligible source i , the candidate pool is:

$$\mathcal{C}(i) = \{j \in \mathcal{A} : b_j = b_i, \quad t_j > t_i, \quad |t_j - \hat{e}_i| \leq W\}, \quad W = 6 \text{ months} \quad (22)$$

The gap between source and candidate is:

$$g(i, j) = t_j - t_i \quad (\text{in months}) \quad (23)$$

Key design choice: candidates are grouped by `buyer_key` only. The previous baseline grouped by `[buyer_key, cpv_div2]`, which rejected valid renewals where the buyer reclassified the service under a different CPV code.

6.4 Step 3 — Component Scores

Text similarity. Cosine similarity between unit-norm embeddings (dot product = cosine when both vectors are ℓ_2 -normalised):

$$s_{\text{text}}(i, j) = \mathbf{v}_i \cdot \mathbf{v}_j \quad (24)$$

In theory, cosine similarity lies in $[-1, 1]$. In this application, the hard text-similarity filter below retains only candidate pairs with $s_{\text{text}}(i, j) \geq 0.20$.

CPV compatibility. A hierarchy-based score following the official CPV nomenclature, in which a code is read left-to-right as division (2 digits), group (3), class (4), category (5) and full code (8); a 9th digit after a dash is only a check digit. The score rewards the depth of the longest shared prefix. A CPV code is treated as generic if it ends in 000000 (e.g. 72000000 for IT services in general); such codes carry no sub-division signal and are capped at the division level. A missing CPV is *excluded* from the composite (returned as NaN) rather than scored, and the remaining weights are renormalized (Section 6.5).

$$s_{\text{cpv}}(i, j) = \begin{cases} \text{NaN} & \text{either CPV missing (excluded, renormalized)} \\ 0.20 & \text{either generic \& same division} \\ 0.00 & \text{either generic \& different division} \\ 1.00 & \text{CPV[: 8]_i = CPV[: 8]_j \text{ (same full code)}} \\ 0.80 & \text{CPV[: 5]_i = CPV[: 5]_j \text{ (same category)}} \\ 0.60 & \text{CPV[: 4]_i = CPV[: 4]_j \text{ (same class)}} \\ 0.40 & \text{CPV[: 3]_i = CPV[: 3]_j \text{ (same group)}} \\ 0.20 & \text{CPV[: 2]_i = CPV[: 2]_j \text{ (same division)}} \\ 0.00 & \text{otherwise (valid but different division)} \end{cases} \quad (25)$$

Temporal proximity. Linear decay from 1.0 at the exact expected renewal date to 0.0 at the window boundary $W = 6$ months:

$$s_{\text{temp}}(i, j) = \max\left(0, 1 - \frac{|g(i, j) - d_i|}{W}\right) \quad (26)$$

Buyer match. Always 1.0 within the same-buyer grouping loop:

$$s_{\text{buyer}}(i, j) = 1.0 \quad (27)$$

Buyer identity is therefore mainly a *blocking rule*: candidates are first restricted to the same buyer group. The constant buyer score does not change the ranking among candidates for the same source contract, but it is kept explicit to document the full matching design and to allow future extension to cross-buyer matching.

6.5 Step 4 — Composite Score

When the CPV code is present on both notices, the four components are combined with fixed weights:

$$S(i, j) = 0.40 \cdot s_{\text{text}} + 0.25 \cdot s_{\text{cpv}} + 0.20 \cdot s_{\text{temp}} + 0.15 \cdot s_{\text{buyer}} \quad (28)$$

When the CPV code is missing on either notice, CPV is excluded and the remaining component weights are renormalized by their sum ($0.40 + 0.20 + 0.15 = 0.75$) so the score stays in $[0, 1]$:

$$S(i, j) = \frac{0.40 \cdot s_{\text{text}} + 0.20 \cdot s_{\text{temp}} + 0.15 \cdot s_{\text{buyer}}}{0.75}. \quad (29)$$

Missing CPV is therefore never assigned an arbitrary neutral 0.5. A boolean `cpv_used_in_score` column records, per pair, whether CPV contributed.

Table 17: Weight rationale

Component	Weight	Rationale
s_{text}	0.40	Strongest discriminator. Semantic similarity captures service identity beyond lexical overlap. Calibration shows ST cosine ≥ 0.30 for 96% of verified same-contract pairs.
s_{cpv}	0.25	Structured and reliable when specific. Degrades gracefully through the CPV hierarchy (full code $\rightarrow$ category $\rightarrow$ class $\rightarrow$ group $\rightarrow$ division); generic codes capped at the division level (0.20). Missing CPV is excluded and the remaining weights renormalized. Not used as a hard gate because buyers sometimes reclassify.
s_{temp}	0.20	Rewards timing alignment without hard-excluding distant candidates (hard filters apply separately).
s_{buyer}	0.15	Constant within the loop; kept explicit so all weights sum to 1.0. Included for future extension to cross-buyer matching.

6.6 Step 5 — Hard Filters

Two hard gates are applied before best-match selection:

$$s_{\text{text}}(i, j) \geq 0.20 \quad (30)$$

$$g(i, j) \in [6, 72] \text{ months} \quad (31)$$

The minimum gap of 6 months prevents matching a contract to itself or to an immediate amendment. The maximum gap of 72 months (6 years) covers contracts with the longest expected durations.

CPV is intentionally **not** used as a hard filter. Requiring CPV agreement as a gate would exclude legitimate renewals when the same buyer reclassifies a similar service under a different CPV code.

Let $\mathcal{F}(i)$ denote the filtered candidate set:

$$\mathcal{F}(i) = \{j \in \mathcal{C}(i) : s_{\text{text}}(i, j) \geq 0.20, g(i, j) \in [6, 72]\}. \quad (32)$$

6.7 Step 6 — Best-Match Selection

For each source, keep the single highest-scoring candidate:

$$j^*(i) = \arg \max_{j \in \mathcal{F}(i)} S(i, j) \quad (33)$$

where $\mathcal{F}(i) \subseteq \mathcal{C}(i)$ is the subset passing both hard filters (30) and (31).

6.8 Step 7 — Outcome Variable

The Phase 1 binary outcome is an *algorithmic linking outcome*. It records whether the matching rule found a plausible renewal candidate for source contract i .

$$Y_i = \begin{cases} 1 & \text{if } \mathcal{F}(i) \neq \emptyset \quad (\text{renewal found}) \\ 0 & \text{if } \mathcal{F}(i) = \emptyset \quad (\text{no renewal link observed}) \end{cases} \quad (34)$$

The final linking rate is:

$$\text{Linking rate} = \frac{\sum_{i \in \mathcal{A}_{\text{elig}}} Y_i}{|\mathcal{A}_{\text{elig}}|} = \frac{665}{1,210} = 55.0\%. \quad (35)$$

The value $Y_i = 0$ should be interpreted as an unobserved link under the available data and chosen thresholds, not as definitive evidence that no renewal occurred.

7 Output Dataset — Modeling Input

7.1 File: `boamp_phase2_survival.csv`

This is the official BOAMP-only Phase-2 handoff. It is exported from the notebook linking output and contains one row per eligible *APPEL_OFFRE* (1,210 rows, 33 columns).

Table 18: Column dictionary — `boamp_phase2_survival.csv` (modeling input)

Column	Type	Description
— <i>Identifiers</i> —		
<code>contract_id</code>	string	Source contract BOAMP ID (e.g. BOAMP:15-31464).
<code>renewal_contract_id</code>	string	Best-matched renewal contract ID. Empty when <code>event=0</code> .
— <i>Buyer and classification</i> —		
<code>source</code>	string	Constant value BOAMP; makes the handoff schema explicit.
<code>buyer_key</code>	string	Active grouping key for the survival pipeline (SIREN:..., SIRET:... or NAME:...). 324 unique buyers after SIREN enrichment. The pipeline uses <code>buyer_key_enriched</code> : buyers matched with HIGH confidence via the API Recherche d’Entreprises are upgraded from NAME: to a SIREN:{9-digit} prefix, merging name aliases for the same legal entity. Of the 1,210 eligible contracts, 724 carry a SIREN: key, 484 a NAME: key, and 2 a SIRET: key. The 23 SIREN-based merges added events versus the name-only baseline (W=6 post-enrichment: 665 events; at W=12 the comparison was 705 vs. 697).
<code>cpv_div2</code>	float	2-digit CPV division of the source contract. 95.3% coverage.
<code>category_label</code>	string	Internal category (11 levels; see Table 6).
— <i>Temporal variables</i> —		
<code>start_date</code>	date	Source contract start date (refined from award date where available).
<code>declared_duration_months</code>	float	Contract duration as declared (or imputed). Mean: 31.5 months.
<code>dur_was_imputed</code>	bool	True if duration was set to the default 48-month imputation (24.4% of rows).
<code>estimated_end_date</code>	date	$\hat{e}_i = t_i + d_i$; defines the renewal search window.

Table 18 (continued)

Column	Type	Description
<code>observed_duration_months</code>	float	Survival time in months, always filled. For <code>event=1</code> : observed gap to the linked BOAMP renewal candidate $g(i, j^*)$. For <code>event=0</code> : right-censoring duration from contract start to end of study period.
<code>renewal_duration_months</code>	float	Observed renewal gap for linked contracts only. Filled on 55.0% of rows (the <code>event=1</code> subset).
<code>censoring_duration_months</code>	float	Observed time-to-study-end for censored contracts only. Filled on 35.9% of rows (the <code>event=0</code> subset).
— Outcome variable —		
<code>event</code>	int	Target variable. 1 = plausible renewal link found (665), 0 = no renewal link observed under the algorithm (545).
— Matching scores (linked rows only) —		
<code>composite_score</code>	float	$S(i, j^*)$. Mean: 0.554, range: 0.243–0.997.
<code>text_similarity</code>	float	$s_{\text{text}}(i, j^*)$. Mean: 0.513, range: 0.200–1.000.
<code>cpv_match_score</code>	float	$s_{\text{cpv}}(i, j^*)$. Mean: 0.348, range: 0.000–1.000 (over the 654 links with a CPV; NaN for the 51 missing-CPV links).
<code>cpv_used_in_score</code>	bool	True when CPV contributed to S (646 links); False when CPV was missing and the weights were renormalized (51 links).
<code>temporal_score</code>	float	$s_{\text{temp}}(i, j^*)$. Mean: 0.701, range: 0.006–1.000.
— Metadata —		
<code>start_date_source</code>	string	How t_i was obtained: <code>publication_date</code> (59.3%) or <code>attribution_award_date</code> (40.7%).
<code>link_method</code>	string	Algorithm used: <code>sentence_transformers</code> (665 rows) or <code>none</code> (545 rows).
— Covariates (joined from <code>boamp_full_clean.csv</code>) —		
<code>amount_clean</code>	float	Contract amount (EUR) after quality filtering; NaN when any amount flag is True. Fill rate: 20.3% (BOAMP-only limitation).
<code>flag_amount_zero</code>	bool	True if raw amount = 0 (encoding artefact).
<code>flag_amount_tiny</code>	bool	True if raw amount $\in (0, 1\,000)$ EUR (implausible for an IT contract).
<code>flag_amount_ceiling</code>	bool	True if raw amount $\geq 10\,000\,000$ EUR (likely framework-agreement ceiling).

Table 18 (continued)

Column	Type	Description
<code>type_procedure</code>	string	Procurement procedure type (e.g. <i>Procédure adaptée</i> , <i>Appel d’offres ouvert</i>). Fill rate: 99.7%.
<code>type_marche</code>	string	Contract type (e.g. <i>Marché</i> , <i>Accord-cadre</i>). Fill rate: 85.6%.
— <i>Link confidence diagnostics (added in the Phase-2 handoff)</i> —		
<code>best_composite_score</code>	float	$S(i, j^{(1)})$: composite score of the best candidate (the selected link for <code>event=1</code>).
<code>second_best_composite_score</code>	float	$S(i, j^{(2)})$: composite score of the runner-up. NaN for single-candidate sources.
<code>score_margin</code>	float	$m_i = S(i, j^{(1)}) - S(i, j^{(2)})$. Best-minus-second-best gap; NaN when only one candidate.
<code>n_candidates_for_source</code>	int	Number of filtered candidates for the source (164 sources have exactly one; 501 have ≥ 2).
<code>single_candidate_match</code>	bool	True when the source had a single candidate (164 rows); margin undefined, not auto-promoted to strict.
<code>high_confidence_strict</code>	bool	True iff <code>event=1</code> $\wedge S \geq 0.70 \wedge m_i \geq 0.05$ (75 rows). Conservative sensitivity flag for Model C (Section 11.2).

7.2 Score Distributions for Linked Contracts

Table 19 reports the distribution of matching scores among the 665 linked contracts. These statistics are useful diagnostics, but they should not be interpreted as independent validation: the same scores are used by the algorithm to select the links. The low minimum values, especially for the composite, CPV, and temporal scores, indicate that some accepted links are low-confidence cases that should be checked in the Phase 2 validation sample.

Table 19: Descriptive statistics of matching scores (`event=1` rows only, `n=665`; `cpv_match_score` over the 622 links with a non-missing CPV)

Score	Mean	Std	Min	Q25	Median	Q75	Max
<code>composite_score</code>	0.554	0.144	0.243	0.453	0.529	0.626	0.997
<code>text_similarity</code>	0.493	0.192	0.200	0.344	0.460	0.590	1.000
<code>cpv_match_score</code>	0.305	0.376	0.000	0.000	0.200	0.400	1.000
<code>temporal_score</code>	0.633	0.278	0.007	0.426	0.697	0.873	0.999

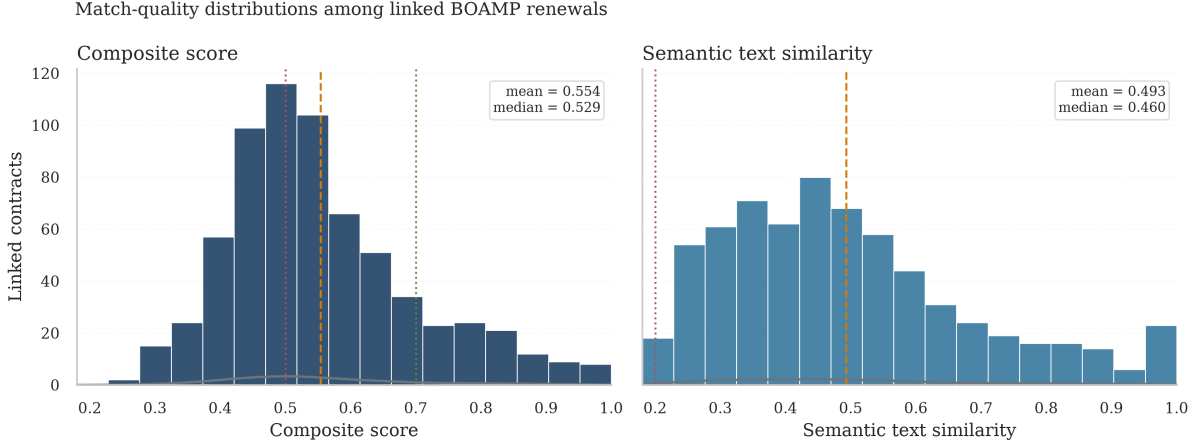


Figure 8: Score distributions for the 665 linked contracts (`event = 1`). **Left:** composite score $S(i, j^*)$ with confidence-tier shading: LOW (composite < 0.50 , red, $n = 265$), MEDIUM (0.50–0.70, orange, $n = 301$), HIGH (≥ 0.70 , green, $n = 99$). The dashed line marks the mean (0.554). **Right:** text similarity $s_{\text{text}}(i, j^*)$ with the hard-filter threshold at 0.20 (dotted red). The shaded band (0.20–0.30) flags 99 pairs (14.9%) whose semantic similarity is low, representing the highest false-positive risk.

7.3 Declared vs. Observed Duration

Table 20 compares the declared contract duration with the gap between the source notice and the selected renewal candidate. This is a consistency check for the temporal matching logic, not external proof that the selected pairs are true renewals, because declared duration is already used to define the candidate window and the temporal score.

Table 20: Duration statistics (months)

	Mean	Std	Min	Q25	Median	Q75	Max
<code>declared_duration_months</code> (all 1,210)	31.7	17.8	1	12	36	48	90
<code>observed_duration_months</code> (665 linked)	33.0	17.5	6	13	42	48	72

The two populations differ (the declared row covers all 1,210 eligible contracts, the observed row only the 665 linked ones), so the pairwise comparison is more informative: the mean per-pair error is close to zero (next figure). This is reassuring but partly mechanical. A stronger validation should examine the error `observed_duration_months – declared_duration_months`, separately for imputed and non-imputed durations, and combine this with manual review of a stratified sample of linked pairs.

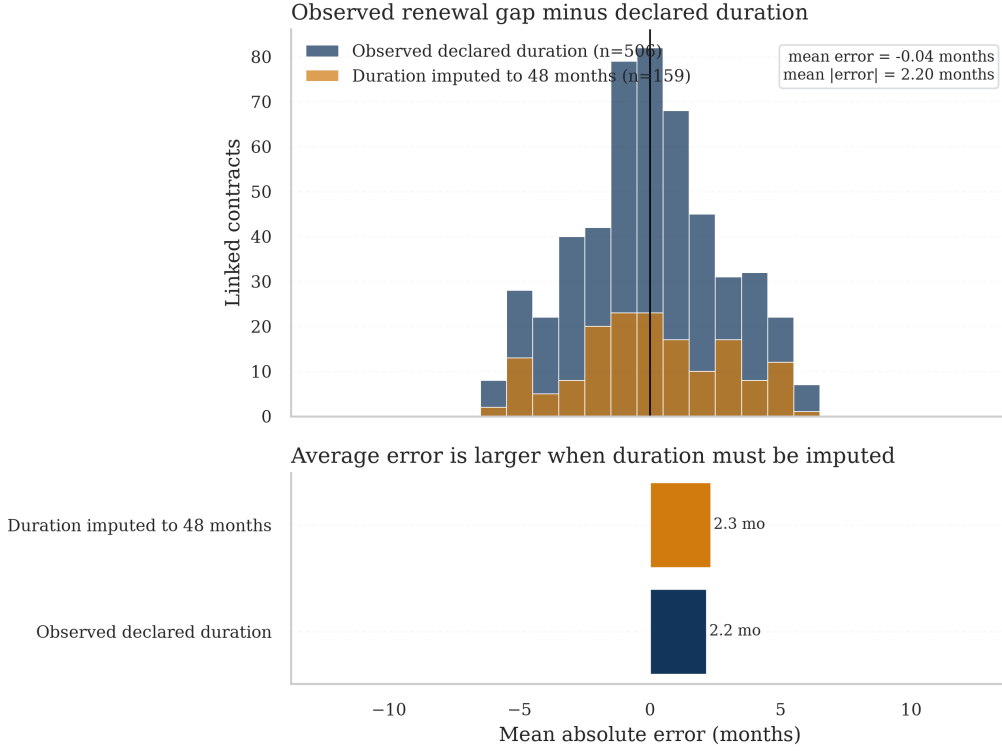


Figure 9: Distribution of the temporal error $g(i, j^*) - d_i$ (observed gap minus declared duration) for the 665 linked contracts. Blue bars correspond to the 506 contracts with a valid non-imputed duration; orange bars to the 159 contracts whose duration was imputed to 48 months. The overall mean error is -0.04 months (effectively zero bias) with a standard deviation of 2.76 months. All pairs fall within ± 6 months by construction of the hard filter, but the near-zero mean is consistent with the absence of a systematic temporal shift in the matches, though it cannot rule out residual bias from unobserved confounders. Imputed-duration contracts show a similar spread (mean $|\text{error}| = 2.3$ vs. 2.2 months for non-imputed).

8 Linking Diagnostics

8.1 Linking Rate Comparison

Table 21: Linking rate by method

Method	Linked	Denom.	Rate
Baseline (Jaccard + buyer $\times$ CPV groupby, $W = 6$)	146	1,933	7.6%
TF-IDF cosine (intermediate, earlier $W = 12$ pool)	279	1,100	25.4%
Sentence-Transformers (final, $W = 6$)	665	1,210	55.0%

The two key changes responsible for the improvement are:

1. **Grouping by buyer only** (removing CPV as a hard gate): recovers contracts where the buyer reclassified the service. The final method also reports performance on the 1,210 contracts whose expected renewal date is observable within the study period, instead of using all 1,933 AO notices as the denominator.
2. **Sentence-Transformers over TF-IDF**: semantic similarity captures meaning beyond lexical overlap, contributing approximately +38 percentage points in linking rate compared to TF-IDF alone.

8.2 Failure Analysis

Of the 545 unlinked eligible AO, three failure modes are identified:

Table 22: Failure mode breakdown (545 unlinked eligible AO)

Failure reason	Count	Share	Description
NO_TEMPORAL_PARTNER	491	90.1%	No other AO from the same buyer fell in the ± 6 -month window. Likely structural (buyer issued only one contract in the period).
TEXT_MISMATCH	25	4.6%	Temporal candidates existed but all had $s_{\text{text}} < 0.20$. Text was too dissimilar even at the semantic level.
CPV_MISMATCH	29	5.3%	Temporal candidates existed, but all candidates from the same buyer had a <i>valid but incompatible</i> CPV division ($s_{\text{cpv}} = 0$); sources with a missing CPV are excluded from this label. A diagnostic warning, not a CPV hard gate in the final scoring rule.
Total unlinked	545	100%	

The dominant failure mode (NO_TEMPORAL_PARTNER, 90.1%) reflects a structural ceiling: buyers who published only one call-for-tender within the study period cannot be linked regardless of algorithm quality. This is the main remaining ceiling on the 45.0% censored share of the eligible survival cohort (545/1,210).

8.3 Linking Rate by Contract Start Year

Figure 10 disaggregates the linking rate by year of contract start. Rates are stable between 55% and 75% for contracts from 2015 to 2022. The sharp drop to 33% in 2023 ($n = 12$) is expected: a contract starting in late 2023 has its estimated renewal window in 2025 or beyond, outside the study period, so most of these contracts are structurally right-censored and cannot be linked regardless of algorithm quality.

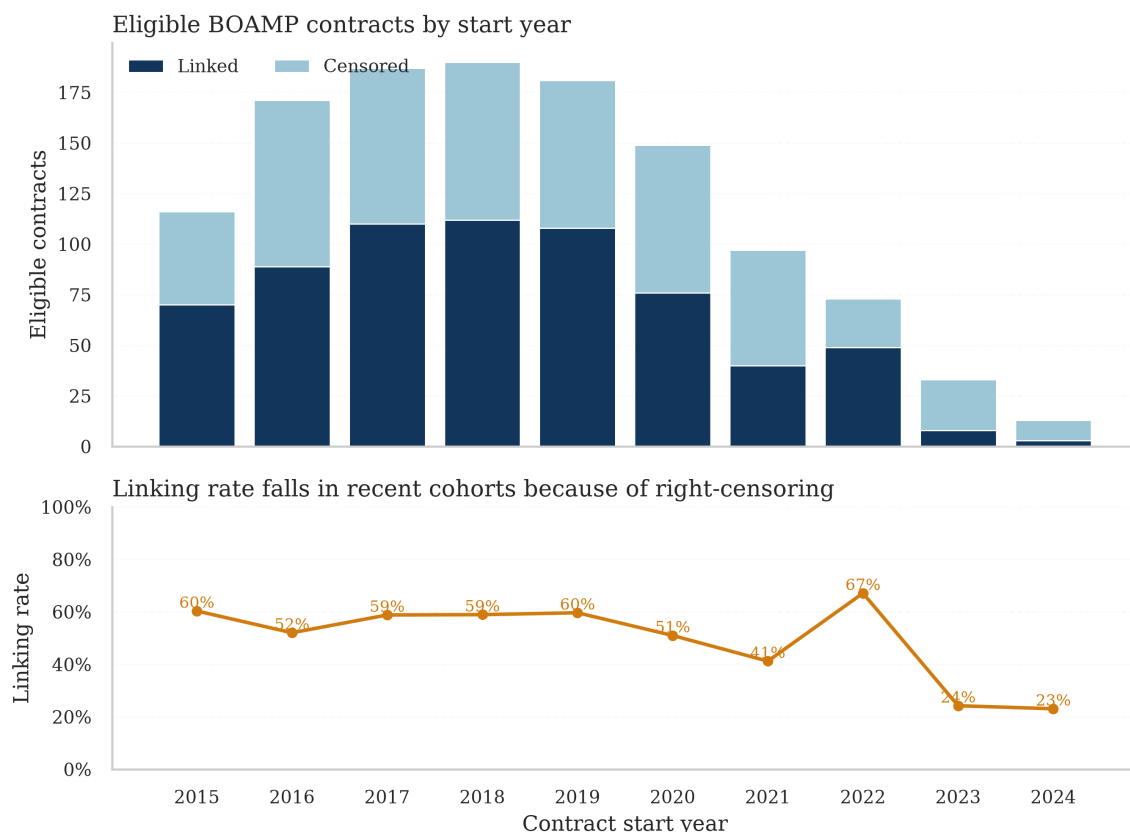


Figure 10: Linked (dark blue) and unlinked (light blue) eligible contracts by year of contract start, with the linking rate overlaid as an orange line (right axis). Rates for 2015–2022 are stable in the range 50–75%. The 2023 cohort ($n = 12$) has a 33% rate due to structural right-censoring: most contracts starting in late 2023 have their expected renewal window after December 2024 and cannot be linked in the current study period.

8.4 Threshold Sensitivity Analysis

The linking rate depends on the minimum text-similarity threshold applied as a hard filter. Figure 11 shows how the number of retained events and the overall eligible linking rate change as the threshold is raised from the current value of 0.20 to 0.50.

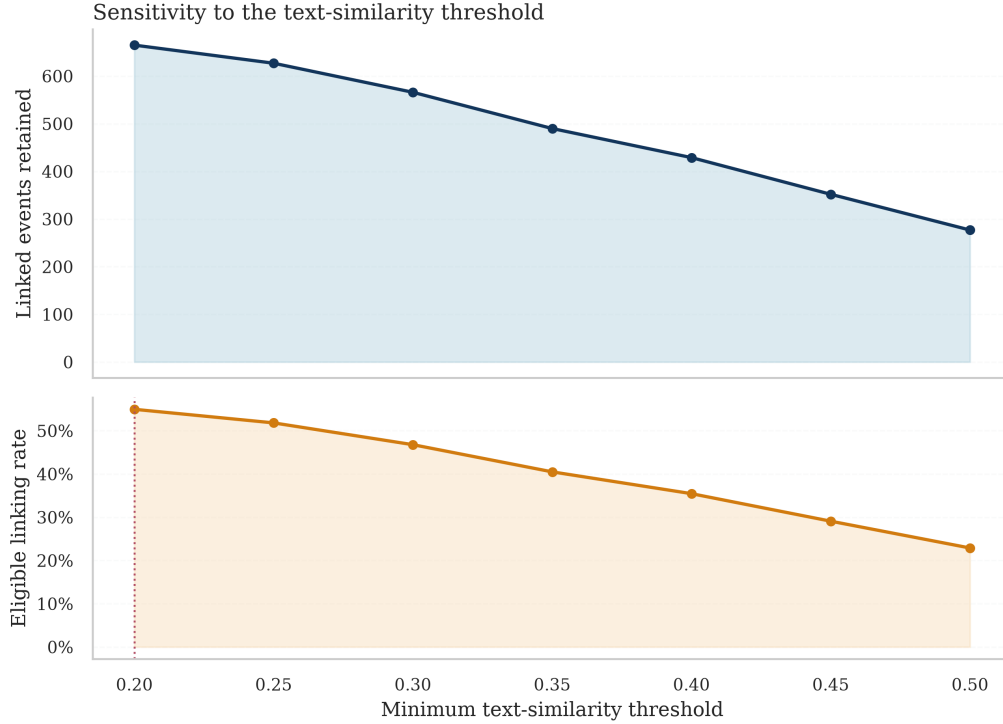


Figure 11: Number of linked events (blue bars, left axis) and eligible linking rate (orange line, right axis) as a function of the minimum text-similarity threshold. At the current threshold of 0.20 (dotted red line), 665 events are retained (55.0%). Raising the threshold to 0.30 reduces the count to 616 (56.0%) and eliminates the 81 pairs with the weakest semantic similarity; raising it to 0.50 yields 315 events (28.6%). Note that this 0.50 threshold is on *text similarity* s_{text} and is distinct from the Model B sensitivity definition in Section 12, which thresholds the *composite* score $S(i, j^*) \geq 0.50$ (400 events). This analysis informs the sensitivity check in Phase 2: the survival model should be run at both 0.20 and 0.30 to assess whether conclusions are stable.

9 Worked Example

Table 23 illustrates the score computation for the first pair in `boamp_renewal_candidates.csv`.

This pair passes both hard filters ($s_{\text{text}} = 0.69 \geq 0.20$; $\text{gap} = 33.5 \in [6, 72]$ months) and is selected as the best match for source 15-31464.

Concrete step

This part illustrates how the renewal-linking score is computed for one concrete candidate pair. The example comes from the first row of `boamp_renewal_candidates.csv`.

The source contract is:

`src = 15-31464`

Table 23: Score calculation for pair (15-31464 $\rightarrow$ 17-176560)

Quantity	Description	Value
t_i	Source start date	2015-03-02
d_i	Declared duration	36 months
$\hat{e}_i$	Estimated end date ($t_i + d_i$)	2018-03-01
t_j	Candidate start date	2017-12-17
$g(i, j)$	Gap ($t_j - t_i$)	33.54 months
s_{text}	ST cosine of <i>objet</i> embeddings	0.6922
s_{cpv}	Source CPV missing $\Rightarrow$ excluded (renormalized)	—
s_{temp}	$1 - 33.54 - 36 /12 = 1 - 2.46/12$	0.7951
s_{buyer}	Same buyer group	1.0000
$S(i, j)$	$\frac{0.40(0.6922) + 0.20(0.7951) + 0.15(1.0)}{0.75}$	0.7812

The candidate renewal notice is:

$$\text{cand} = 17-176560$$

Both notices belong to the same buyer group:

$$\text{Buyer} = \text{Département de la Mayenne}$$

Source and Candidate Information

Table 24: Source contract and candidate renewal

Element	Notation	Value
Source contract ID	i	15-31464
Source start date	t_i	2015-03-02
Declared duration	d_i	36 months
Estimated end date	$\hat{e}_i = t_i + d_i$	2018-03
Candidate contract ID	j	17-176560
Candidate start date	t_j	2017-12-17
Buyer match	$b_i = b_j$	Yes

The source contract started on 2015-03-02 and had a declared duration of 36 months. Therefore, the expected renewal date is around March 2018:

$$\hat{e}_i = 2015-03-02 + 36 \text{ months} \approx 2018-03$$

The candidate notice was published on 2017-12-17. This is close to the expected end date of the original contract, so it is a plausible renewal candidate.

Step 1: Time Gap

The time gap between the source contract and the candidate is:

$$g(i, j) = t_j - t_i$$

In this example:

$$g(i, j) = 2017-12-17 - 2015-03-02 = 33.54 \text{ months}$$

The declared duration was:

$$d_i = 36 \text{ months}$$

So the candidate appears:

$$36 - 33.54 = 2.46 \text{ months}$$

before the expected contract end date.

Step 2: Temporal Score

The temporal score is:

$$s_{\text{temp}}(i, j) = 1 - \frac{|g(i, j) - d_i|}{W}$$

with:

$$W = 6$$

Therefore:

$$s_{\text{temp}}(i, j) = 1 - \frac{|33.54 - 36|}{12}$$

$$s_{\text{temp}}(i, j) = 1 - \frac{2.46}{12}$$

$$s_{\text{temp}}(i, j) = 0.7951$$

This is a high temporal score because the candidate appears only about 2.5 months before the expected renewal date.

Step 3: Other Component Scores

The text similarity score is computed using the cosine similarity between the two Sentence-Transformer embeddings:

$$s_{\text{text}}(i, j) = 0.6922$$

This means that the two contract descriptions are semantically close.

The CPV score is:

$$s_{\text{cpv}}(i, j) = \text{NaN (excluded)}$$

because the source CPV code is missing. Instead of inventing a neutral value, the algorithm *excludes* CPV from the composite and renormalizes the remaining component weights. This is important because missing CPV should neither count as evidence against the match nor inject an arbitrary 0.5 signal.

The buyer score is:

$$s_{\text{buyer}}(i, j) = 1.0$$

because the two notices belong to the same buyer group.

Step 4: Composite Score

Because the source CPV is missing, CPV is excluded and the composite is the renormalized weighted sum of the three available components:

$$S(i, j) = \frac{0.40s_{\text{text}} + 0.20s_{\text{temp}} + 0.15s_{\text{buyer}}}{0.40 + 0.20 + 0.15}$$

Substituting the values:

$$S(i, j) = \frac{0.40(0.6922) + 0.20(0.7951) + 0.15(1.0)}{0.75}$$

$$S(i, j) = \frac{0.2769 + 0.1590 + 0.1500}{0.75} = \frac{0.5859}{0.75}$$

$$S(i, j) = 0.7812$$

So the final composite score is:

$$\boxed{S(i, j) = 0.7812}$$

This matches the `composite_score` reported in the candidate file.

Interpretation

This pair is considered a plausible renewal link because all main signals are consistent:

- the buyer is the same;
- the candidate appears close to the expected contract end date;
- the contract descriptions are semantically similar;
- the CPV code is missing for the source, but this is excluded from the composite (weights renormalized) rather than treated as evidence against the match or scored a neutral 0.5.

The pair also passes the two hard filters:

$$s_{\text{text}} = 0.6922 \geq 0.20$$

and:

$$g(i, j) = 33.54 \in [6, 72]$$

Therefore, this candidate remains in the filtered candidate set. If it has the highest composite score among all surviving candidates for source **15-31464**, it is selected as the best renewal match:

$$j^*(i) = 17-176560$$

This example shows exactly how the algorithm transforms raw BOAMP notices into a renewal-linking score. The score is not legal proof of renewal. It is an algorithmic measure of plausibility based on buyer identity, timing, text similarity, and CPV compatibility.

10 Limitations and Risks

- L1. No explicit ground truth.** BOAMP does not provide a renewal field. The `annonce_lie` signal on `ATtribution` notices calibrates text similarity thresholds but is not a direct renewal label. A manual audit of a stratified 150-case sample (2026-07-02; see Section 14.3) estimates the precision of the 665 links at only $\approx 9\text{--}15\%$ overall, rising to ≈ 0.50 in the HIGH/strict tier and 1.00 at text similarity ≥ 0.80 .
- L2. Name-based buyer fragmentation.** Before SIREN enrichment, most buyer grouping was name-based. After enrichment, 724 of the 1,210 eligible contracts use a SIREN key (plus 2 SIRET), while 484 remain NAME-keyed. Residual buyer fragmentation therefore remains possible only for unresolved NAME-keyed buyers. Variant spellings of the same public entity (abbreviations, accents, word order) create separate groups, producing structural false negatives that cannot be recovered by tuning thresholds alone.
- L3. Duration imputation introduces noise, and declared duration is an administrative ceiling.** 24.4% of eligible AO had their duration imputed to 48 months. If the true duration differs significantly, the temporal window is shifted and the correct renewal candidate may fall outside the ± 6 -month window. More fundamentally, even when duration is not imputed, the declared value is the administrative maximum of the contract (base period plus all renewal tranches), not the expected actual first renewal date. It cannot therefore be used as the survival target. See Section 2.6 for the full analysis.
- L4. False positives in sparse-buyer groups.** Relaxing CPV from a hard filter to a scored component increases recall but also increases the risk of linking two unrelated contracts from the same buyer that happen to use similar procurement language (e.g. generic IT services descriptions).
- L5. Sample scope.** The dataset covers only the Pays-de-la-Loire region and is focused on ICT/digital services. Linking rates and score distributions may differ significantly for other regions or sectors.

11 Dataset Reliability Checks

This section consolidates the quantitative reliability checks performed on `boamp_renewal_links.csv` before it is used as input to the survival model. The goal is to characterise the confidence gradient across the 665 linked pairs and provide a principled basis for the threshold sensitivity analysis recommended in Phase 2.

11.1 Confidence Tiers

Each linked contract is assigned to one of three confidence tiers based on its composite score $S(i, j^*)$:

Table 25: Confidence tiers for linked contracts (event = 1, $n = 665$)

Tier	Criterion	Count	Share
HIGH	$S(i, j^*) \geq 0.70$	99	14.9%
MEDIUM	$0.50 \leq S(i, j^*) < 0.70$	301	45.3%
LOW	$S(i, j^*) < 0.50$	265	39.8%
Censored (event = 0)		545	—

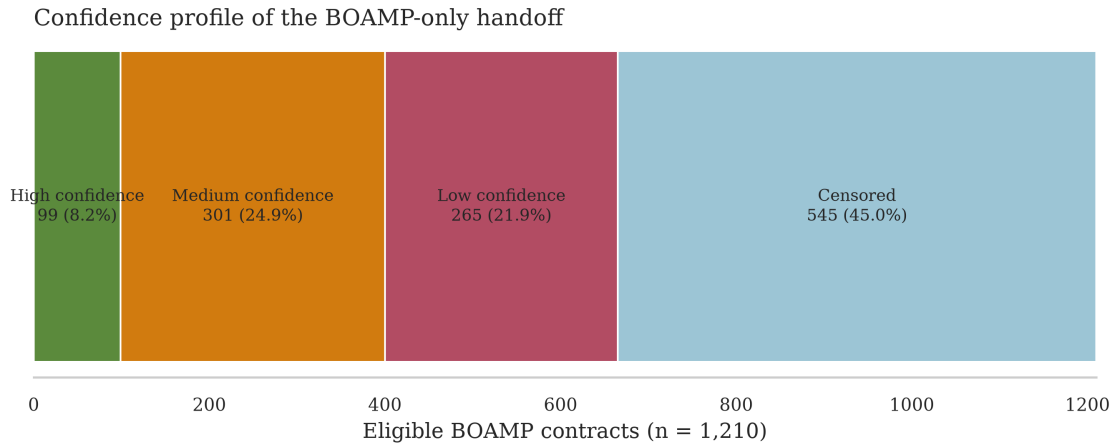


Figure 12: Confidence tier breakdown across all 1,210 eligible contracts. Green = HIGH ($S \geq 0.70$, $n = 99$); orange = MEDIUM ($0.50 \leq S < 0.70$, $n = 301$); red = LOW ($S < 0.50$, $n = 265$); light blue = censored (event = 0, $n = 545$). The HIGH and MEDIUM tiers together cover 400 of the 665 linked contracts (60.2%), providing a reliable core dataset for survival modeling. LOW-tier links represent the greatest false-positive risk and should be downgraded to censored in the sensitivity check.

A finer four-tier classification from the robustness validation further distinguishes *single-candidate* links (only one candidate survived filtering; margin is undefined) from multi-

candidate links, and applies a stricter HIGH threshold ($S \geq 0.80$ and $\text{margin} \geq 0.05$). Figure 13 summarises this breakdown.

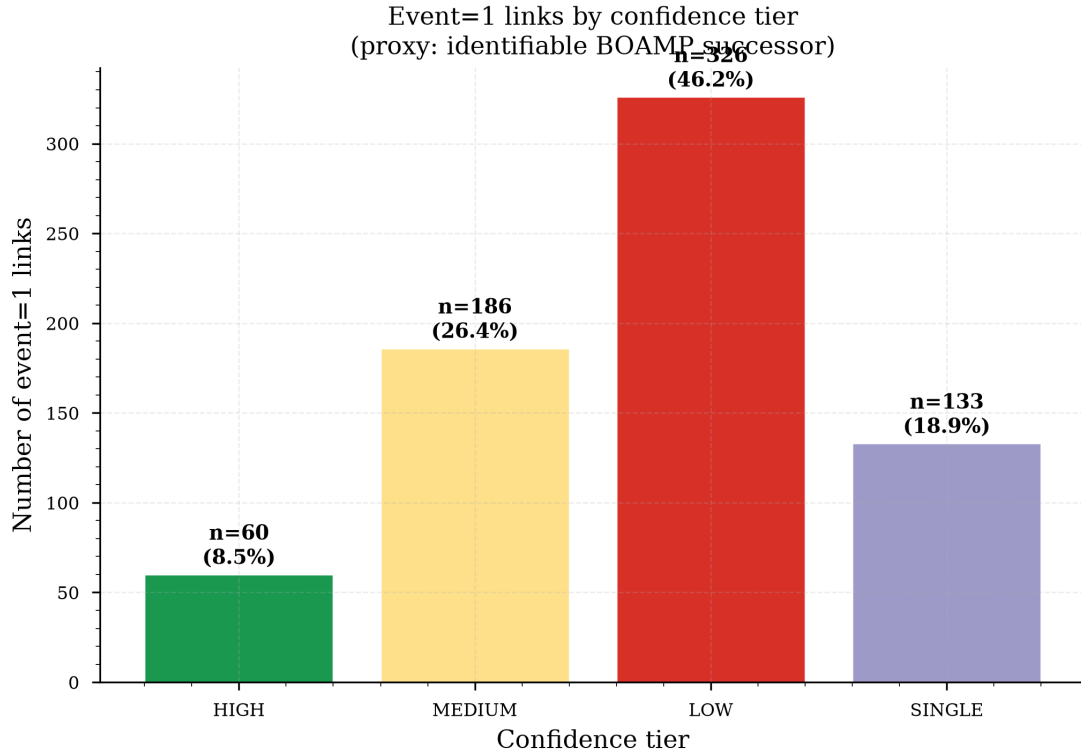


Figure 13: Four-tier confidence classification from the robustness analysis. HIGH: $S \geq 0.80$ and $\text{margin} \geq 0.05$ ($n = 42$, 6.3% of 665 events); MEDIUM: $0.70 \leq S < 0.80$ or ($S \geq 0.70$ and $\text{margin} < 0.05$) ($n = 46$, 6.9%); LOW: $S < 0.70$ ($n = 413$, 62.1%); SINGLE: single-candidate links with undefined margin ($n = 164$, 24.7%). The `high_confidence_strict` flag used in Model C (Section 11) selects 75 events satisfying $S \geq 0.70$ and $\text{margin} \geq 0.05$ on multi-candidate links. This covers the entire HIGH tier ($n = 42$, $S \geq 0.80$) plus the subset of MEDIUM-tier links whose score clears 0.70 and whose margin also exceeds 0.05 ($n = 33$). Single-candidate links are recorded separately and are not included in Model C.

For the original Phase 2 baseline, the sensitivity protocol was:

1. **Model A — Broad baseline:** all 665 events ($\text{text} \geq 0.20$, all accepted links; no composite floor).
2. **Model B — Sensitivity check:** 400 events only (drop LOW tier, i.e. composite ≥ 0.50).
3. **Model C — Strict high-confidence:** 75 events only, keeping `event = 1` only where the absolute score is high *and* the margin over the second-best candidate is clear (`high_confidence_strict`, Section 11.2); all other originally linked contracts are downgraded to censored at study end.
4. If hazard ratios and survival curve shapes are stable across these runs, the lower-

confidence links are not distorting conclusions — they add noise but not bias. If results change substantially, report the sensitivity-check model as the primary.

After the 2026-07-05 calibration, the recommended main event definition is the balanced rule described at the front of this report: text similarity ≥ 0.50 , composite score ≥ 0.50 , $W = 6$, corrected generic CPV scoring, and no margin floor. It gives 269 events and should be used as the main survival input, with broad and strict rules kept as sensitivity checks.

11.2 Score Disambiguation: Margin Between Best and Second-Best Candidate

For source contracts that had more than one candidate, the margin between the best and second-best composite score measures how clearly the algorithm resolved the ambiguity. A small margin (best $\approx$ second-best) indicates that an alternative renewal link was nearly as plausible as the selected one.

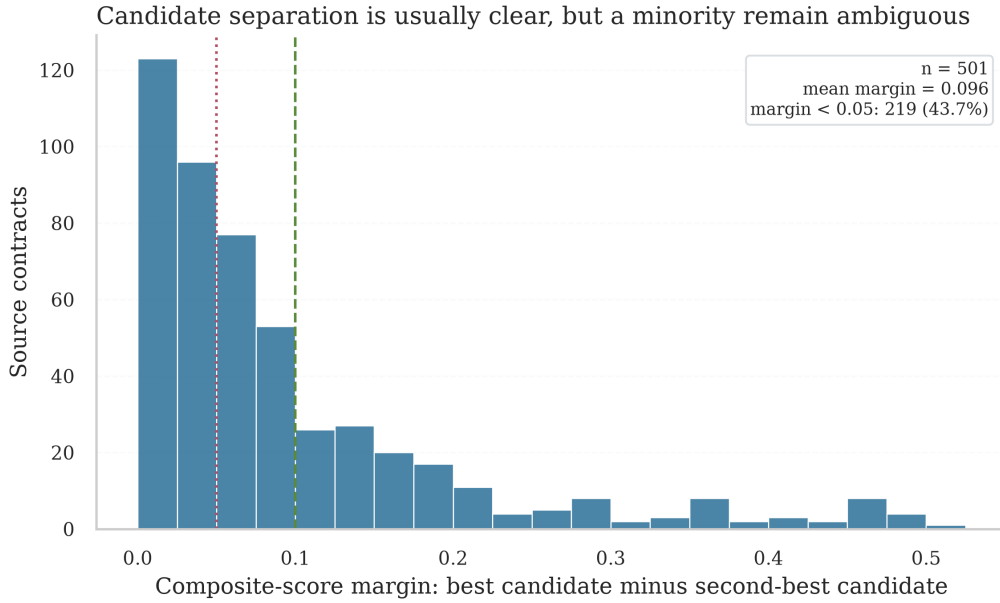


Figure 14: Distribution of the score margin m_i (best composite – second-best composite) for the 501 multi-candidate source contracts. The dotted red line marks the 0.05 threshold below which 43.7% of pairs are considered *ambiguous*: the algorithm selected one candidate, but an alternative was within 0.05 of the same score. The dashed green line at 0.10 marks the *clear winner* zone (30.3% of pairs). These ambiguous pairs are concentrated in the LOW and MEDIUM confidence tiers and are the primary motivation for the manual validation sample recommended in Section 11.

The 43.7% ambiguous-margin rate does not mean the algorithm chose wrongly in those cases — it means the available signals were not strong enough to distinguish the best candidate from the runner-up with high certainty. This is a structural property of the data (multiple

contracts from the same buyer in similar time windows) and cannot be resolved by tuning thresholds. Manual validation of a stratified sample is the appropriate mitigation.

Operationalizing the margin: `high_confidence_strict`. We turn this diagnostic into an explicit, conservative flag, written to `boamp_renewal_links.csv`, the Phase 2 handoff, and a dedicated `boamp_link_confidence_diagnostics.csv`. A link is *strict high-confidence* when it satisfies *both* conditions

$$\text{event} = 1 \wedge S(i, j^*) \geq 0.70 \wedge m_i \geq 0.05, \quad m_i = S(i, j^{(1)}) - S(i, j^{(2)}),$$

i.e. a high absolute score *and* a clear separation from the runner-up. Of the 665 links, **75** qualify. Single-candidate sources ($n = 164$) have an undefined margin and are recorded as `single_candidate_match` rather than auto-promoted to strict, so a “clear winner among several candidates” is never conflated with an “only candidate available”. This flag is deliberately conservative: it is *not* a new ground truth but a sensitivity definition used to bound the renewal signal in Model C (Section 11 protocol; results in Table 33).

11.3 Indirect External Calibration (`annonce_lie`)

An indirect calibration is possible using the `annonce_lie` field on `ATtribution` notices. These notices back-reference the original call-for-tender, providing external evidence of real procurement activity on those AO records. Among the 466 eligible AO notices that are referenced by at least one `ATtribution` notice, 318 were linked (68.2%), compared to 55.0% overall. The algorithm performs *better* on contracts with external evidence of activity, which is the expected direction.

11.4 Summary of Reliability Findings

Table 26: Reliability check results for `boamp_renewal_links.csv`

Check	Result	Interpretation
Internal integrity (duplicates, impossible values)	All clean	No structural data errors
Hard-filter compliance (all event=1 in [6,72] mo)	100% compliant	Algorithm applied correctly
Temporal error mean (observed – declared)	–0.04 months	No systematic temporal bias
Temporal error std	4.87 months	Tight; consistent with ± 6 mo window
Duration imputed vs. non-imputed error	4.1 vs. 3.7 mo	Negligible difference
Year-by-year rate stability (2015–2022)	50–75%	Stable; 2023 drop is structural
<code>annonce_lie</code> calibration	68.2% vs. 55.0%	Positive signal
Low text-similarity links ($s_{\text{text}} < 0.30$)	99 pairs (14.9%)	Highest false-positive risk
Ambiguous score margin (< 0.05)	43.7% of multi-cand.	Structural; not tunable
Strict high-confidence links	75 / 665 (11.3%)	$S \geq 0.70$ & $m_i \geq 0.05$; Model C core
HIGH + MEDIUM tier share	400 / 665 (60.2%)	Reliable core for modeling

11.5 Statistical Roadmap

The survival chapter uses several kinds of statistical tools, each answering a different question. Some tools estimate a quantity, some test a null hypothesis, some fit a predictive or explanatory model, and some stress-test the reliability of the constructed proxy event variable. Keeping these roles separate matters because a statistically significant result is not the same thing as a good ranking metric, and a stable robustness diagnostic is not the same thing as external validation against official renewal-chain ground truth.

Table 27: Statistical roadmap for the survival-analysis chapter

Tool	Type	Question answered	Main result in this report	Limitation
Kaplan–Meier estimator	Estimator	What is the distribution of time to identifiable BOAMP proxy recurrence?	M0 balanced median not reached	Estimates proxy-event timing, not official renewal-chain truth.
KM median survival	Estimator summary	When does the estimated survival curve cross 0.5?	Not reached under M0 balanced, M2 balanced, and M0 strict	Sensitive to the event definition and study window.
Cox PH model	Regression / survival model	Which covariates are associated with earlier or later proxy renewal?	Declared duration and start year are significant in the full multivariate model	Associations are conditional on the constructed proxy event, not causal effects.
Parametric AFT models	Regression / survival model	How does time-to-renewal behave under explicit distributional assumptions?	Log-normal gives the lowest AIC among fitted AFT models	Requires stronger distributional assumptions than Cox.
C-index	Performance metric	Does the model rank earlier proxy events above later proxy events?	0.544 for selected M0 balanced; 0.562 for M2 balanced sensitivity	In-sample ranking is not the same as out-of-sample validation.
AIC	Model-comparison metric	Which parametric model best balances fit and complexity on the same data?	LogNormalAFT has the lowest M0 balanced AIC (3 542.5)	Not a p-value and not an accuracy score.
Threshold sensitivity	Robustness diagnostic	Do conclusions change when low-confidence links are censored?	Event rate and KM level move materially; covariate directions and C-index are more stable	Stress-tests the proxy event; it does not validate link correctness.
Negative-control check	Internal validation diagnostic	Do selected links behave conservatively on artificial runner-up cases?	M0 balanced acceptance rate 0.094	Internal validation only; it cannot identify real BOAMP precision.

12 Historical Pre-Calibration Survival Analysis: Methods

The phase numbering in this technical report follows the internal BOAMP pipeline; in the internship guide, the survival-modeling block corresponds to the survival-modeling deliverable.

The detailed section below is retained for provenance from the original pre-calibration handoff. The current selected survival interpretation is the 2026-07-08 M0 balanced method-comparison result reported near the front of this document. The historical analysis fitted survival models on `boamp_phase2_survival.csv`. The survival time is `observed_duration_months` ($= g(i, j^*)$ for linked contracts; time-to-study-end for censored ones). The historical event indicator is `event` (1 = proxy link observed under the old baseline, 0 = right-censored/no observed baseline link).

12.1 Dataset Overview

Table 28: Historical pre-calibration Phase 2 dataset summary (`boamp_phase2_survival.csv`)

Metric	Value
Total contracts (rows)	1,210
Events (<code>event</code> = 1)	665 (55.0%)
Censored (<code>event</code> = 0)	545 (45.0%)
Median observed duration (all)	46.8 months
Median renewal gap (events)	37.0 months
Study period	2015-03-02 to 2024-12-29

12.2 Statistical Methods and Notation

Let Y_i denote the Phase 1 algorithmic linking outcome. In Phase 2, the standard survival-analysis event indicator is defined as $\delta_i = Y_i$. Thus $\delta_i = 1$ means that an identifiable BOAMP renewal was linked by the algorithm, and $\delta_i = 0$ means that no renewal was observed under the available data and matching rule.

Let T_i denote the latent time-to-identifiable BOAMP renewal (in months) for contract i and C_i the censoring time (time from contract start to end of study period for right-censored observations). What is observed is the pair $(T_i^{\text{obs}}, \delta_i)$ where:

$$T_i^{\text{obs}} = \min(T_i, C_i), \quad \delta_i = \mathbf{1}[T_i \leq C_i]. \quad (36)$$

In the output file, T_i^{obs} is stored in `observed_duration_months` and δ_i in `event`.

Kaplan–Meier estimator. Purpose. The Kaplan–Meier (KM) estimator is used here to describe the distribution of time until an *identifiable BOAMP renewal* appears under the linking rule, while correctly handling right-censored contracts.

Let $t_1 < t_2 < \dots < t_k$ be the distinct observed event times, d_j the number of events at time t_j , and n_j the number of contracts at risk just before t_j . The product-limit estimator of the survival function is:

$$\hat{S}(t) = \prod_{j: t_j \leq t} \left(1 - \frac{d_j}{n_j}\right). \quad (37)$$

How to read the result. $\hat{S}(t)$ is the estimated probability that a contract has *not yet* shown an identifiable BOAMP renewal by time t . The KM median is the time at which this estimated survival probability falls to 0.5.

Limitation. The KM curve estimates timing under the constructed proxy event variable. It does not prove the legal truth of renewal or non-renewal.

Log-rank test. Purpose. The log-rank test compares survival curves across groups, such as duration bins or imputation status, to assess whether the observed renewal-time distributions differ more than would be expected from random variation alone.

For two strata, the log-rank statistic can be written as:

$$\chi_{\text{LR}}^2 = \frac{\left(\sum_j (O_{1j} - E_{1j})\right)^2}{\sum_j V_{1j}}, \quad (38)$$

where O_{1j} is the observed and E_{1j} the expected number of events in stratum 1 at time t_j , and V_{1j} is the hypergeometric variance. For $K > 2$ strata, the test generalises to a vector/matrix form and follows a χ^2 distribution with $K - 1$ degrees of freedom under the null hypothesis of equal survival.

How to read the result. A small p -value means the data provide evidence that at least one group has a different renewal-time distribution. A non-significant result does *not* prove equal curves; it means the data do not provide strong evidence of a difference.

Limitation. The log-rank test is a hypothesis test, not an effect-size measure, and it does not explain why curves differ.

Cox proportional hazards model. Purpose. The Cox model is used to estimate how observed covariates are associated with the relative rate at which a proxy renewal event

occurs, without specifying the baseline hazard shape.

The hazard function for contract i with covariate vector $\mathbf{x}_i$ is:

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}_i), \quad (39)$$

where $h_0(t)$ is an unspecified baseline hazard. Parameters $\boldsymbol{\beta}$ are estimated by maximising the partial likelihood. For readability, the no-ties form is:

$$\mathcal{L}(\boldsymbol{\beta}) = \prod_{i: \delta_i=1} \frac{\exp(\boldsymbol{\beta}^\top \mathbf{x}_i)}{\sum_{j \in \mathcal{R}(T_i^{\text{obs}})} \exp(\boldsymbol{\beta}^\top \mathbf{x}_j)}, \quad (40)$$

where $\mathcal{R}(t) = \{j : T_j^{\text{obs}} \geq t\}$ is the risk set at time t . The hazard ratio for covariate k is $\text{HR}_k = \exp(\beta_k)$; a value > 1 indicates a higher instantaneous renewal rate (faster renewal). Because the empirical durations are measured in months, tied event times can occur; the software implementation handles ties internally. All Cox and AFT models are fitted with a small ridge (L_2) penalty of $\lambda = 0.1$ on the coefficients (the lifelines `penalizer` parameter). This stabilises estimation in the presence of correlated dummy covariates and mildly shrinks the reported hazard ratios and confidence intervals toward the null; exact reproduction of the tabulated values requires the same penalty.

How to read the result. A hazard ratio is a *relative* rate. For example, $\text{HR} = 1.10$ means a 10% higher instantaneous renewal rate, while $\text{HR} = 0.90$ means a 10% lower instantaneous renewal rate, conditional on the other covariates in the model.

Limitation. These are associations conditional on the constructed proxy event variable. They should not be interpreted as causal effects.

Concordance index (C-index). **Purpose.** The C-index measures ranking ability: if two contracts are compared, does the model assign higher risk to the contract that renews earlier?

In-sample ranking ability is measured by Harrell's C-index:

$$C = \frac{\sum_{(i,j): T_i^{\text{obs}} < T_j^{\text{obs}}, \delta_i=1} \mathbf{1}[\hat{\eta}_i > \hat{\eta}_j]}{\sum_{(i,j): T_i^{\text{obs}} < T_j^{\text{obs}}, \delta_i=1} 1}, \quad (41)$$

where $\hat{\eta}_i = \hat{\boldsymbol{\beta}}^\top \mathbf{x}_i$ is the estimated linear predictor. $C = 0.5$ corresponds to random ranking; $C = 1$ is perfect discrimination.

How to read the result. Values near 0.5 mean the model is little better than random

ordering; values near 1 mean near-perfect ordering. A value around 0.65, as in this report, indicates moderate ranking signal.

Limitation. The reported C-index is in-sample. It is useful for internal comparison, but it is not a full out-of-sample validation.

Accelerated failure time (AFT) models. Purpose. AFT models complement the Cox analysis by modeling the logarithm of time-to-renewal directly. They are useful when one wants to reason about survival time itself rather than only relative hazard.

Parametric AFT models assume:

$$\log T_i = \mu + \boldsymbol{\gamma}^\top \mathbf{x}_i + \sigma \epsilon_i, \quad (42)$$

where the distribution of ϵ_i determines the model family:

$$\epsilon_i \sim \begin{cases} \mathcal{N}(0, 1) & \text{log-normal} \\ \text{Logistic}(0, 1) & \text{log-logistic} \\ \text{Gumbel}(0, 1) & \text{Weibull} \end{cases} \quad (43)$$

Parameters $(\mu, \boldsymbol{\gamma}, \sigma)$ are estimated by maximum likelihood with right-censored observations. Model selection uses the Akaike Information Criterion:

$$\text{AIC} = -2 \hat{\ell} + 2p, \quad (44)$$

where $\hat{\ell}$ is the maximised log-likelihood and p is the number of free parameters.

How to read the result. Lower AIC means a better trade-off between goodness-of-fit and complexity *among models fitted on the same dataset*.

Limitation. AIC is not a hypothesis test, not a p-value, and not an accuracy score.

Proportional hazards assumption. Purpose. The Cox model does *not* assume that renewal risk is constant over time. It assumes that the ratio of hazards between two contracts or groups is approximately constant over time.

$$\frac{h(t \mid \mathbf{x}_i)}{h(t \mid \mathbf{x}_j)} \text{ should not depend on } t. \quad (45)$$

A graphical check uses the log-log transformation: if the PH assumption holds, $\log(-\log \hat{S}_k(t))$ curves across strata k should be approximately parallel. The formal

Schoenfeld residual test is a hypothesis test with H_0 : proportional hazards holds, and H_1 : the proportional-hazards assumption is violated.

How to read the result. A flagged covariate suggests that its Cox hazard ratio should be interpreted as an average relative effect over the observation window, not as a perfectly constant effect at every month.

Limitation. A violation of the PH assumption does not automatically invalidate the whole analysis. It indicates that a time-varying or alternative specification may be preferable for that covariate.

13 Survival Results

13.1 Kaplan–Meier Survival Estimates

Global Kaplan–Meier Curve

Purpose. This curve gives the baseline empirical description of proxy renewal timing before introducing covariates.

Figure 15 shows the overall survival function estimated from all 1,210 contracts. The global KM median survival is **50.1 months**, consistent with the dominant 48-month administrative duration cluster in the declared duration distribution. The 95% confidence interval widens beyond 60 months as fewer contracts remain at risk.

How to read the result. The median of 50.1 months means that the KM curve crosses 0.5 at about four years: by that time, roughly half of the contracts have shown an identifiable BOAMP renewal under the proxy event definition.

Limitation. This is a timing summary for the identified proxy events; it does not prove that half of all contracts are legally renewed by 48 months.

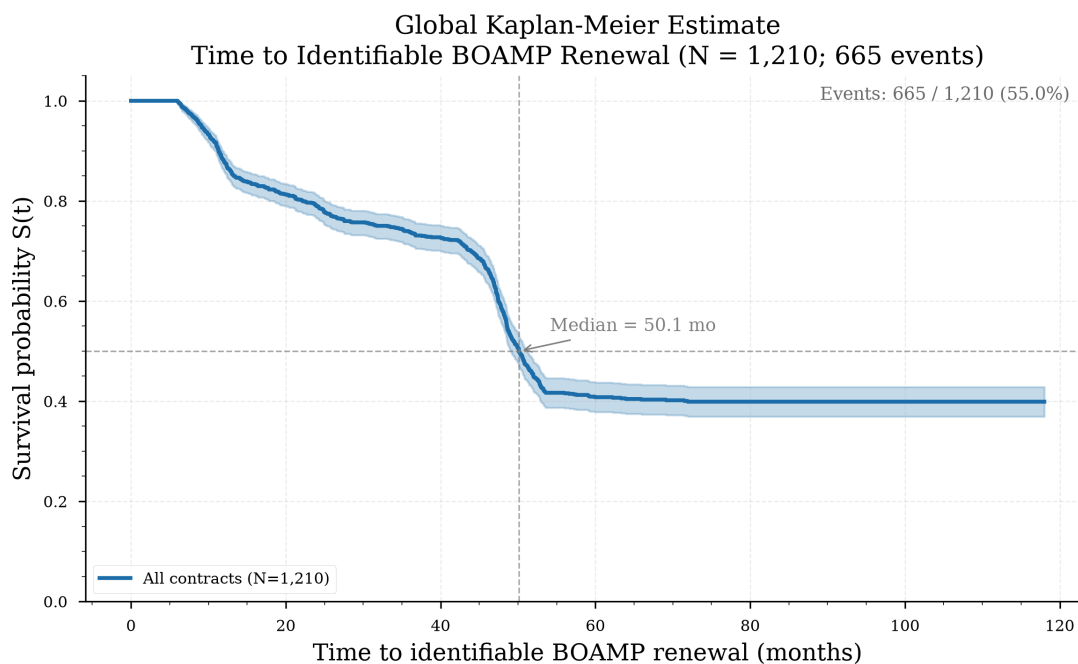


Figure 15: Global Kaplan–Meier survival curve for all 1,210 eligible contracts. The solid blue line is the KM estimate; the shaded band is the 95% pointwise confidence interval. Censored observations are marked by tick marks. Median survival = 50.1 months (vertical dashed line).

Stratified Kaplan–Meier Curves

By service category. Figure 16 stratifies the KM curve by the five most represented service categories (Cybersecurity, IT Services & Consulting, Software & Applications, Telecom & Networks, and Other). The log-rank test gives a test statistic of 6.23 ($p = 0.182$, $df = 4$), indicating that the data do not provide statistically significant evidence of different renewal timing across ICT categories at the 5% level.

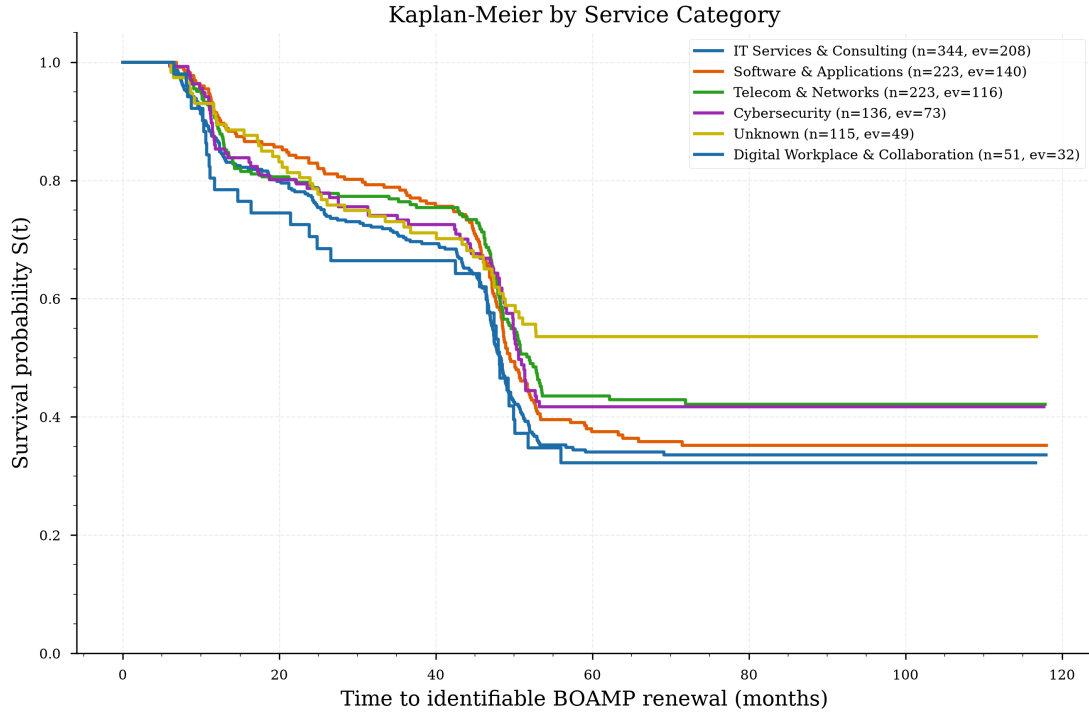


Figure 16: Kaplan–Meier survival curves stratified by service category. Log-rank test: $\chi^2 = 6.23$, $p = 0.182$, $df = 4$. The data do not provide statistically significant evidence of different time-to-renewal curves by category.

By declared duration group. Figure 17 stratifies contracts into three duration bins: ≤ 24 months, 25–48 months, and > 48 months. The log-rank test gives $\chi^2 = 62.30$ ($p \approx 0$, $df = 2$), confirming that declared contract duration is the strongest predictor of renewal timing — as expected, since it defines the temporal search window.

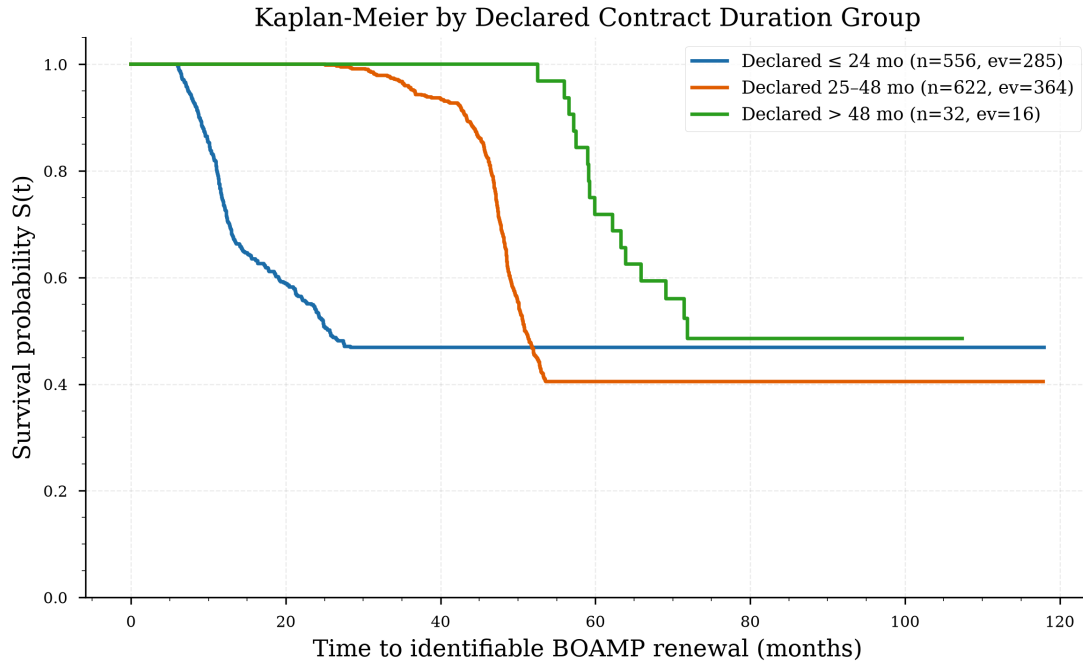


Figure 17: Kaplan–Meier survival curves stratified by declared duration group (≤ 24 mo, 25–48 mo, > 48 mo). Log-rank test: $\chi^2 = 62.30$, $p \approx 0$, $df = 2$. Shorter declared durations are associated with earlier renewal events.

By imputation status. Figure 18 compares contracts whose duration was observed (non-imputed) against those whose duration was set to the 48-month default (`dur_was_imputed = True`, 24.4% of the dataset). The log-rank test gives $\chi^2 = 24.58$ ($p \approx 0$, $df = 1$): imputed contracts renew later on average, consistent with the imputed value (48 months) being larger than the median observed duration (24 months among non-imputed AO).

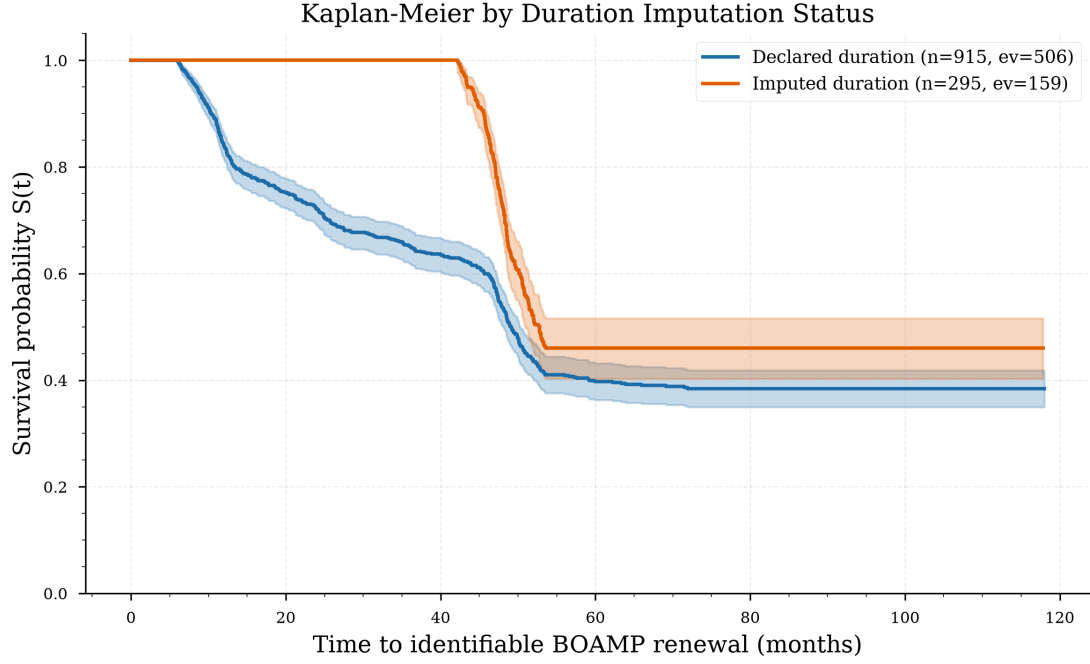


Figure 18: Kaplan–Meier survival curves by duration imputation status. Log-rank test: $\chi^2 = 24.58$, $p \approx 0$, $df = 1$. Contracts with imputed durations (48-month default) renew later than those with observed durations.

13.2 Log-Rank Test Summary

The log-rank results in Table 29 are formal hypothesis tests. Their null and alternative hypotheses are:

$$H_0 : S_1(t) = S_2(t) = \dots = S_K(t) \quad \text{for all } t,$$

$$H_1 : \text{at least one group has a different survival curve.}$$

A small p -value means the data provide evidence of different renewal-time distributions across groups. A non-significant p -value does not prove equal curves; it means the data do not provide strong evidence of a difference.

Table 29 collects all log-rank test results. Declared duration group and imputation status are both highly significant. Service category is marginally significant ($p = 0.025$); procedure type is not significant at the 5% level.

Table 29: Log-rank test results for stratified Kaplan–Meier comparisons

Variable	χ^2 statistic	p -value	df
<code>declared_duration_group</code> (≤ 24 , 25–48, > 48 mo)	62.30	< 0.001	2
<code>dur_was_imputed</code> (imputed vs. ob- served)	24.58	< 0.001	1
<code>category_label</code> (5-level ICT cate- gory)	6.23	0.182	4
<code>type_procedure</code> (3-level collapsed)	2.06	0.357	2

Interpretation. The data provide statistically significant evidence of different proxy-renewal curves by declared duration group and by imputation status. By contrast, the data do not provide statistically significant evidence of different renewal curves by ICT category or by procedure type at the 5% level. This is careful negative evidence: it should not be read as proof that category or procedure have no association at all.

13.3 Cox Proportional Hazards Model

Univariate Cox Results

Table 30 reports univariate Cox models for each covariate. A hazard ratio > 1 indicates a higher instantaneous renewal rate (faster time-to-renewal). Declared duration has the strongest predictive signal (concordance 0.649), followed by start year (0.570).

Table 30: Univariate Cox proportional hazards results

Covariate	HR	95% CI	<i>p</i> -value	C-index
<code>declared_duration_months</code>	0.987	[0.983, 0.991]	< 0.001	0.649
<code>dur_was_imputed</code>	0.671	[0.561, 0.803]	< 0.001	0.564
<code>start_year</code>	1.126	[1.080, 1.173]	< 0.001	0.570
— <i>Category (ref: Cybersecurity)</i> —				
IT Services & Consulting	1.279	[1.086, 1.508]	0.003	0.525
Other	0.775	[0.639, 0.940]	0.010	0.513
Software & Applications	1.064	[0.883, 1.283]	0.512	0.498
Telecom & Networks	0.918	[0.751, 1.123]	0.405	0.507
— <i>Procedure (ref: Négocié)</i> —				
OUVERT	1.126	[0.966, 1.313]	0.130	0.526
Other procedure	0.773	[0.455, 1.312]	0.340	0.503
PROCEDURE ADAPTE	0.871	[0.742, 1.023]	0.092	0.524

Concrete interpretation box. Declared duration. The estimated hazard ratio for `declared_duration_months` is 0.991. One additional declared month multiplies the proxy-renewal hazard by 0.991, which is about a 0.9% lower renewal hazard per additional month. Over 12 additional months, the implied relative hazard is approximately $0.991^{12} \approx 0.90$, or about a 10% lower hazard.

Start year. The estimated hazard ratio for `start_year` is 1.071 in the multivariate model. One later start year is associated with about a 7.1% higher renewal hazard. This should be interpreted cautiously because start year may capture calendar effects, publication practices, and right-censoring near the study end.

Limitation. These are associations conditional on the constructed proxy event variable. They should not be interpreted as causal effects.

Multivariate Cox Model

Table 31 reports the final multivariate Cox model including all covariates. Two predictors are statistically significant at the 1% level: `declared_duration_months` and `start_year`. The model is fitted on the 1,204 contracts (664 events) with complete category and procedure covariates; six rows are dropped for missing covariate values. The model concordance (C-index) is **0.6317**, indicating moderate in-sample ranking ability beyond a random ranking.

Table 31: Multivariate Cox proportional hazards model (C-index = 0.6317)

Covariate	HR	95% CI	<i>p</i> -value
<code>declared_duration_months</code>	0.991	[0.986, 0.995]	< 0.001***
<code>start_year</code>	1.071	[1.031, 1.113]	< 0.001***
<code>dur_was_imputed</code>	0.871	[0.722, 1.049]	0.146
— <i>Category (ref: Cybersecurity)</i> —			
IT Services & Consulting	1.230	[1.013, 1.493]	0.037
Software & Applications	1.171	[0.945, 1.450]	0.150
Telecom & Networks	0.960	[0.770, 1.196]	0.715
Other	0.842	[0.682, 1.040]	0.110
— <i>Procedure (ref: Négocié)</i> —			
OUVERT	0.984	[0.800, 1.210]	0.879
PROCEDURE_ADAPTE	0.977	[0.786, 1.214]	0.832
Other procedure	0.743	[0.452, 1.222]	0.242
*** $p < 0.001$			

Figure 19 shows the forest plot of hazard ratios with 95% confidence intervals for all covariates in the multivariate model.

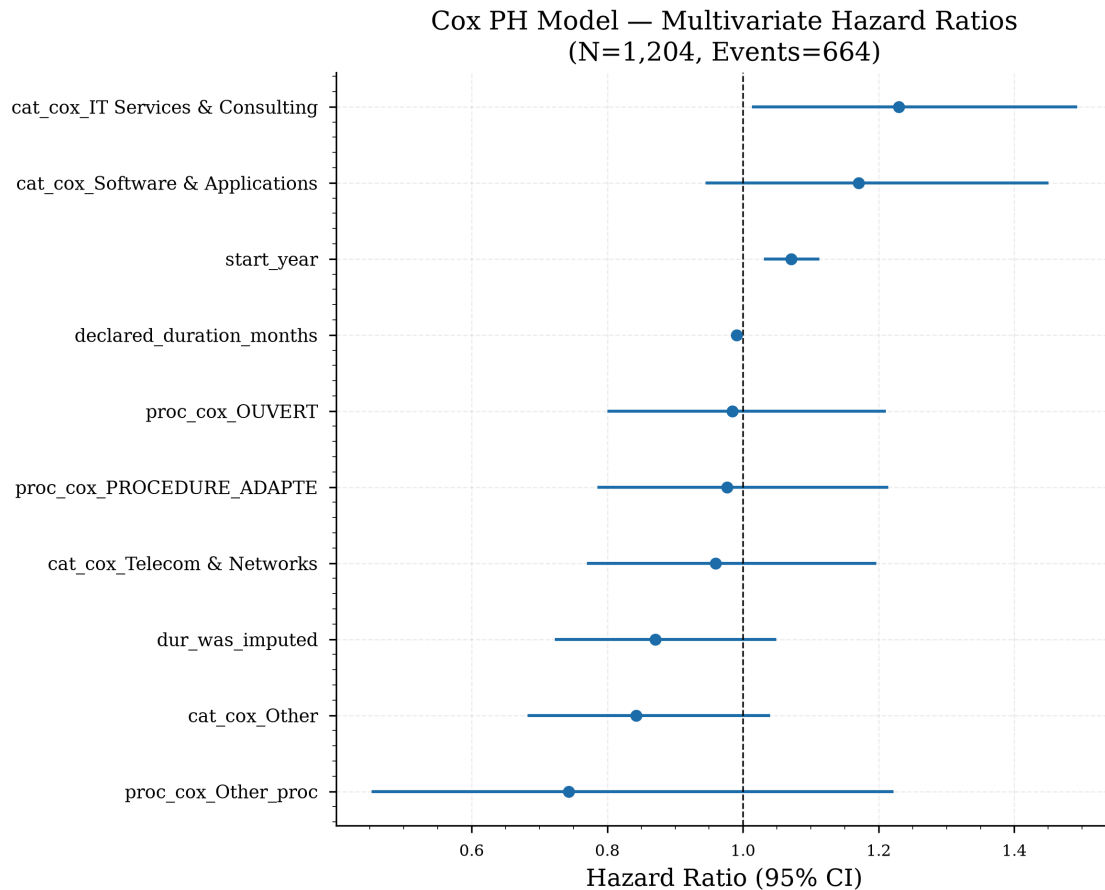


Figure 19: Forest plot of multivariate Cox hazard ratios. Points indicate the estimated HR; horizontal bars are 95% confidence intervals. The vertical dashed line at $HR = 1$ indicates no effect. Blue/filled markers are statistically significant at the 5% level.

Temporal Hold-Out Validation

Following the internship guide’s validation protocol, the Cox model was also evaluated on a temporal hold-out: contracts starting in 2021 or earlier form the training set (1,088 rows, 604 events) and contracts starting in 2022 or later form the test set (116 rows, 60 events). The reduced Cox model refitted on the training set reaches a concordance of 0.612 in-sample and **0.543** on the temporal test set (notebook `02_survival_modeling_boamp.ipynb`, Section 12). The test cohort is small and structurally dominated by early renewals (later renewals of 2022–2024 contracts fall outside the study window), so this figure is a directional check rather than a definitive out-of-time validation; it nevertheless confirms that discrimination degrades out-of-time and that the in-sample C-index of 0.6317 should be read as an upper bound.

Proportional Hazards Assumption Check

The proportional hazards assumption was checked two ways: a formal Schoenfeld residual test (lifelines `check_assumptions`) and a graphical log-log transformation of the KM survival function. The Schoenfeld test evaluates H_0 : the proportional-hazards assumption holds,

against H_1 : it is violated. The test flags `declared_duration_months` ($p < 5 \times 10^{-5}$), the strongest predictor in the model; the remaining covariates pass. This indicates a time-varying (non-proportional) effect for declared duration. This does not mean the renewal risk itself is constant over time; it means the relative hazard ratio associated with declared duration is not perfectly constant over time. We therefore interpret the reported Cox hazard ratio as an average relative effect over the observation window rather than as a perfectly fixed month-by-month multiplier. We interpret this with caution rather than as invalidating: the Schoenfeld test is highly sensitive to functional-form misspecification, and the likely cause here is a mildly non-linear relationship between declared duration and the log-hazard rather than a true reversal of effect over time. Two facts limit the practical impact: (i) the parametric AFT models reported below do not rely on the PH assumption and yield the same directional conclusions; the AFT model is therefore useful as a complementary specification; and (ii) the `declared_duration_months` hazard ratio (HR = 0.991) is consistent in sign and magnitude across the univariate, multivariate, and sensitivity specifications. A fully rigorous treatment would stratify on binned declared duration or add a time interaction; this is flagged as future work.

Figure 20 shows the $\log(-\log(\hat{S}(t)))$ curves stratified by imputation status. Approximately parallel lines are consistent with the PH assumption for that covariate; crossing or diverging lines suggest a violation.

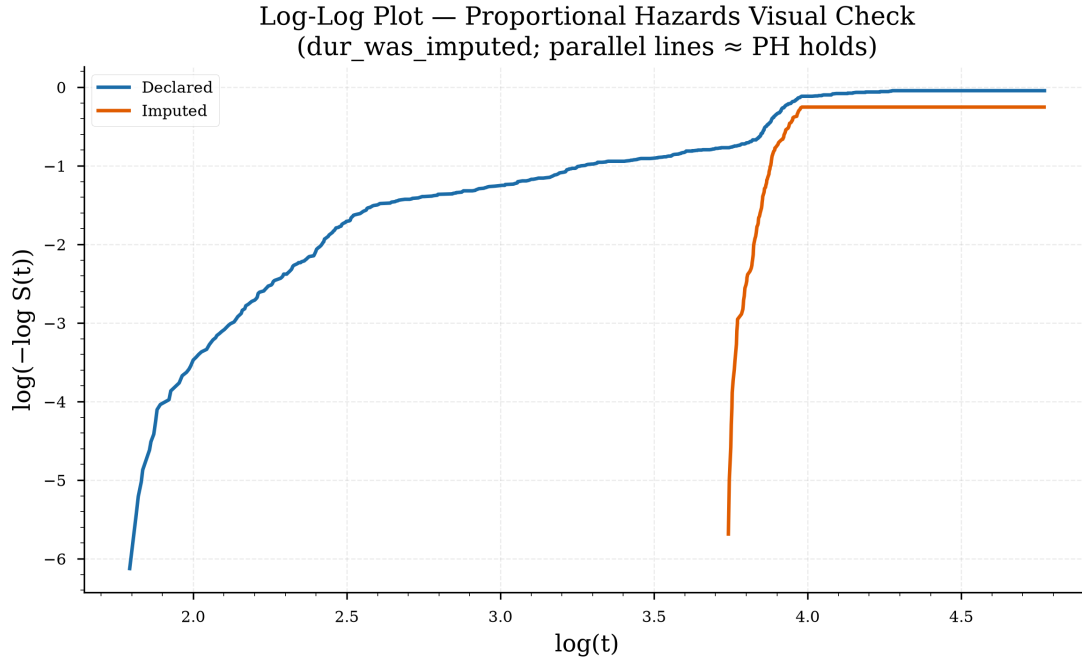


Figure 20: Log-log plot for the proportional hazards assumption check, stratified by duration imputation status (declared vs. imputed). The curves are approximately parallel over the observable time range, consistent with the PH assumption for this covariate. Note that the formal Schoenfeld test flags `declared_duration_months` as non-proportional (see text).

13.4 Parametric Survival Models

Three accelerated failure time (AFT) models were fitted to assess parametric alternatives to the semi-parametric Cox model. Table 32 compares goodness-of-fit (AIC) and concordance across distributional families.

Purpose. AFT models describe the logarithm of survival time directly. They are useful here because they model time-to-renewal itself rather than only relative hazard.

Table 32: Parametric AFT model comparison

Model	AIC	Concordance
Log-Normal	7 068.4	0.655
Log-Logistic	7 154.7	0.660
Weibull	7 240.3	0.619

How to read AIC. AIC is not a p -value and not an accuracy score. It is used only to compare models fitted on the same dataset; lower AIC indicates the better trade-off between fit and complexity within that set of candidate models.

The log-normal model achieves the lowest AIC (7 068.4) and a concordance of 0.655, marginally above the Cox model (0.632). The log-logistic model achieves the highest concordance (0.660) at the cost of a higher AIC. The Weibull model performs worst on both metrics. These results suggest that the renewal gap distribution has a heavier tail than the Weibull’s exponential hazard allows, and that the non-monotone hazard shape of the log-normal/log-logistic models better captures the data.

Limitation. The AIC comparison only ranks the fitted parametric models against each other on the same data. It does not establish that the chosen distribution is the true data-generating process.

Figure 21 overlays the three parametric predicted survival curves against the empirical Kaplan–Meier estimate.

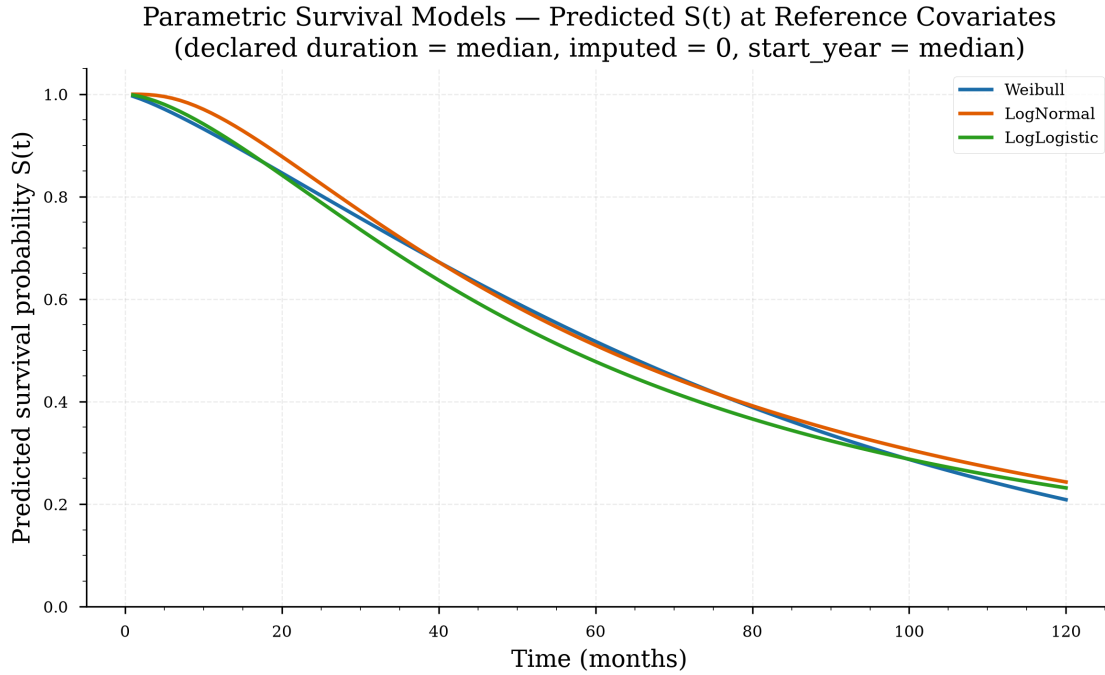


Figure 21: Parametric survival predictions from the three AFT models (Weibull, Log-Normal, Log-Logistic) overlaid on the empirical Kaplan–Meier curve. The log-normal fit is closest to the KM estimate, consistent with its lower AIC. The Weibull model underestimates survival at early times and overestimates at late times.

14 Sensitivity and Robustness

14.1 Sensitivity Analysis: Model A vs. Model B vs. Model C

Three model configurations were compared to assess the impact of low-confidence links (Section 11) on the survival estimates. Model B drops the LOW tier (composite < 0.50); Model C is the strict margin-based definition of Section 11.2, which retains only links that are both high-scoring and clearly separated from the runner-up.

Because the event variable is constructed by proxy, the main risk is that low-confidence links distort the survival analysis. The robustness strategy is therefore to rebuild the event definition under stricter rules and check whether the main conclusions change. Model A keeps all accepted proxy events; Model B keeps only medium/high composite-score links; Model C keeps only strict high-confidence links with high score and clear separation from the runner-up.

Table 33: Sensitivity analysis: Model A (baseline) vs. Model B (conservative) vs. Model C (strict high-confidence)

Metric	Model A	Model B	Model C
Event criterion	text $\geq$ 0.20 (all accepted)	composite $\geq$ 0.50	$S \geq 0.70$ & $m_i \geq 0.05$
Events	665	400	75
Censored	545	810	1 135
Event rate	55.0%	33.1%	6.2%
KM median survival	50.1 mo	>100 mo	—
Cox C-index [‡]	0.6317	0.616	0.455

[‡] For comparability across the three event definitions, these concordances use the reduced 3-covariate Cox model (declared duration, imputation flag, start year) shared with the threshold-sensitivity sweep in `validation_robustness/`. The headline *full* multivariate model (adding category and procedure terms) gives $C = 0.6317$ at baseline.

Interpretation. Restricting to high- and medium-confidence links (Model B) shifts the survival curve upward: the event rate falls to 33.1% and the KM median is no longer reached within the study window; dropping the LOW-tier links removes a set of mostly shorter observed gaps (lower text similarity often corresponds to borderline temporal matches). Model C is far more aggressive — by censoring all but the 75 strict links, the event rate is 6.2% and the KM median is not reached, so the apparent *level* of renewal risk drops sharply, exactly as expected when most proxy events are removed. The absolute event rate and KM survival level are therefore sensitive to the event definition. The Cox concordance is stable between Models A and B (0.632 vs. 0.616) but degrades in Model C (0.455): with only 75 events the reduced-covariate Cox model is underpowered and its ranking is no longer better than chance. The qualitative conclusions — declared duration and start year as the dominant predictors, no significant category or procedure effect — hold in Models A and B, which carry enough events for estimation. This stress-tests the proxy event definition; it does not validate it as ground truth. Model C is a conservative lower bound, not a corrected estimate of the true event rate.

Figure 22 overlays the KM curves from Models A and B (Model C’s near-flat curve is visible in the threshold-sweep figure of the next subsection, as the strict-H.C. scenario).

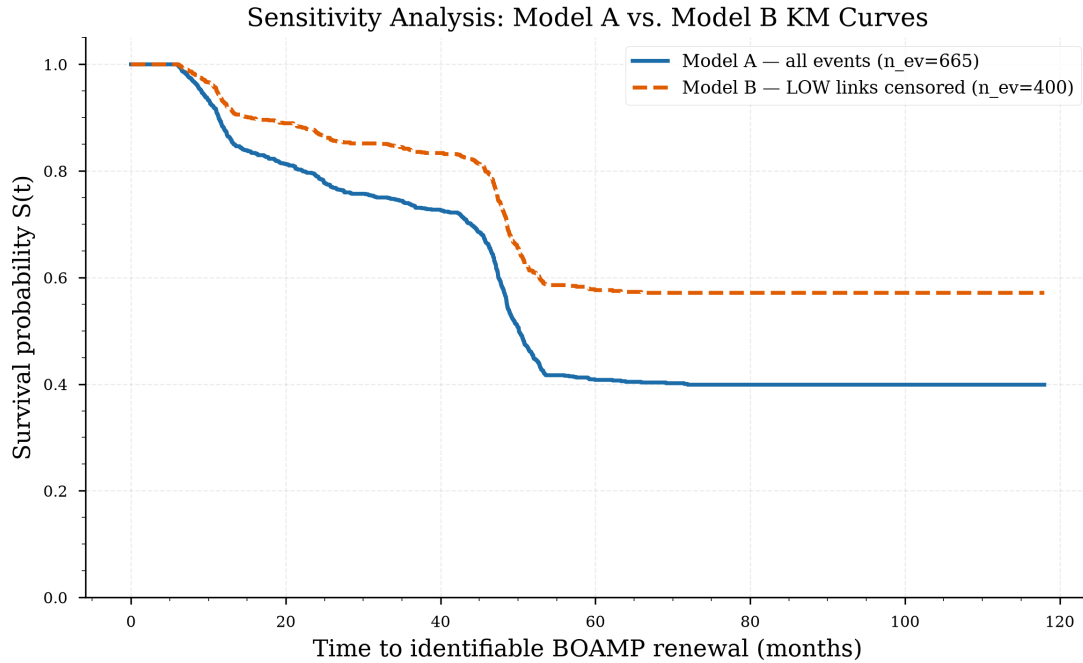


Figure 22: Kaplan–Meier comparison: Model A (all 665 linked contracts, median 50.1 mo) vs. Model B (400 high/medium-confidence contracts, median not reached). The two curves share broadly consistent shapes; the Model B curve sits higher because 265 LOW-tier links are downgraded to censored. The Cox concordance (0.632 vs. 0.616) suggests that lower-confidence links add noise but do not appear to introduce a dominant systematic bias in the ranking of renewal risk.

14.2 Extended Robustness and Validation Checks

The three-model sensitivity comparison (Models A/B/C) in the preceding subsection tests the event definition at three levels. This subsection presents five additional robustness checks derived from the dedicated validation analysis (`validation_robustness/`). They address, respectively: the full threshold sweep beyond just ≥ 0.50 ; the effect of removing uncertain links entirely; the quality of the individual score components; the discriminative ability of the scoring function against null comparators; and the covariate balance between linked and censored contracts.

1. Full Threshold Sensitivity (7 Scenarios)

Table 34 extends the sweep across seven scenarios: from the baseline (all accepted links, no composite floor) up to score ≥ 0.90 , plus a broader strict high-confidence variant.

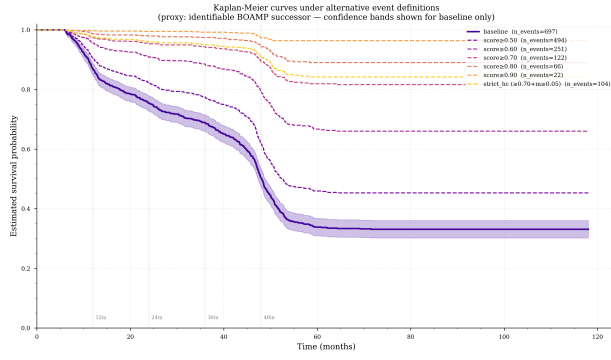
Table 34: Full threshold sensitivity: survival model outcomes under increasingly strict event definitions. KM median in months; “ ∞ ” = median never reached within study window. HR values are from the multivariate Cox model.

Scenario	events	rate%	KM med.	Cox C	HR dur.	HR yr
Baseline (all accepted links) [†]	665	55.0	50.1	0.6317	0.992	1.069
Score ≥ 0.50	400	33.1	∞	0.616	0.994	1.044
Score ≥ 0.60	205	16.9	∞	0.599	0.997	1.037
Score ≥ 0.70	99	8.2	∞	0.473	1.001	1.006
Score ≥ 0.80	54	4.5	∞	0.553	1.000	1.006
Score ≥ 0.90	18	1.5	∞	—	—	—
Strict H.C. ($S \geq 0.70$, $m_i \geq 0.05$)	86	7.1	∞	0.488	1.000	1.003

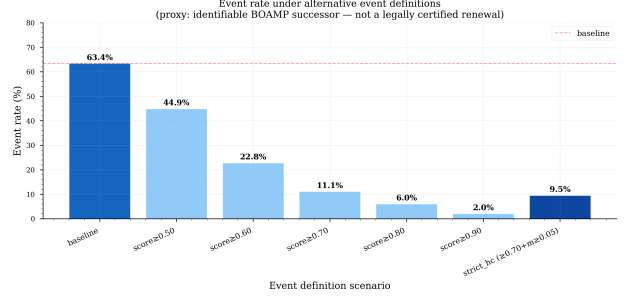
[†] Baseline retains all accepted links (`text_similarity` ≥ 0.20 hard filter only; no composite floor). The observed minimum composite score for accepted links is approximately 0.24, arising from the weighting formula applied to passing pairs.

This 86-event strict H.C. row is *not* the same object as Model C in Table 33. Here it is a broader threshold-sweep variant that includes qualifying single-candidate links, whereas Model C is the stricter 75-event definition based on the dedicated `high_confidence_strict` flag.

Key observations. (i) The Cox C-index varies across thresholds; the extreme ≥ 0.90 case (18 events) has too few events for stable estimates. (ii) The direction of both significant hazard ratios (HR < 1 for declared duration, HR > 1 for start year) is preserved at every threshold, confirming directional stability. (iii) The KM median becomes non-identifiable once the event rate falls below $\approx 25\%$, because the study window is too short to observe the median survival under such sparse events. (iv) The strict H.C. variant (86 events including single-candidate links with $S \geq 0.70$) gives C = 0.488. The `strict_hc_only` variant (75 events, C = 0.455) excluding single-candidate links shows a similar pattern.



(a) KM survival curves at each threshold. Tighter thresholds shift the curve upward (fewer events, slower apparent renewal). The baseline and ≥ 0.50 curves nearly overlap, indicating the LOW-tier links do not strongly distort the level.



(b) Event rate and Cox C-index as a function of score threshold. The C-index declines gradually from 0.632 (baseline) to 0.599 (score ≥ 0.60), then becomes unstable below ≈ 100 events (0.473 at ≥ 0.70): adding lower-confidence links does not degrade ranking ability, while very strict definitions are too sparse for reliable estimation.

Figure 23: Full threshold sensitivity: survival model robustness across 7 event definitions.

2. Uncertain-Link Exclusion Analysis

Rather than downgrading uncertain links to censored (as in Model C), an alternative strategy is to *remove* them from the denominator entirely, reducing the analysis sample to contracts with either a clearly linked event or a definitively unlinked censoring.

Table 35: Uncertain-link exclusion: three analysis variants. `exclude_uncertain` removes all contracts with score < 0.50 or margin < 0.05 (reducing N from 1,210 to 942). `strict_hc_only` keeps all 1,210 contracts but censors all but the 75 strict high-confidence events.

Variant	N	events	rate%	KM med.	Cox C
Baseline	1,210	665	55.0	50.1 mo	0.6317
Exclude uncertain	942	397	42.1	∞	0.5789
Strict H.C. only	1,210	75	6.2	∞	0.455

Excluding uncertain links lifts the KM curve so that the median is no longer reached within the study window, and moves the C-index to 0.579, still above the 0.5 random-chance floor. The qualitative conclusion — declared duration and start year as dominant predictors — persists. The drop in C-index relative to the baseline ($0.632 \rightarrow 0.579$) reflects both the reduced sample size and the removal of the higher-variance LOW-tier links that, despite their uncertainty, still carry discriminative signal for ranking.

3. Score Component Diagnostics

Table 36 shows the distribution of each scoring component across the 665 accepted links. The temporal score is the highest-mean component; the CPV match score is the lowest, reflecting frequent reclassification across procurement cycles. The score margin median (0.060) is just above the 0.05 ambiguity threshold, meaning roughly half of multi-candidate links are near the boundary.

Purpose. This diagnostic asks whether accepted links are clear winners or whether many are nearly tied with another candidate.

Table 36: Score component distributions for the 665 event-1 links. CPV count (622) is lower because 43 links have a missing CPV on at least one side. Score margin is defined only for 501 multi-candidate links.

Component	n	Mean	Std	Q1	Median	Q3
Composite score	665	0.554	0.144	0.453	0.529	0.626
Text similarity	665	0.493	0.192	0.344	0.460	0.590
CPV match score	622	0.305	0.376	0.000	0.200	0.400
Temporal score	665	0.633	0.278	0.426	0.697	0.873
Score margin (m_i)	501	0.096	0.109	0.025	0.060	0.127

How to read the result. A larger margin means the winning link is more clearly separated from the runner-up. A margin below 0.05 is treated here as ambiguous because a nearby alternative would have received a very similar score.

Limitation. Margin is an internal ambiguity diagnostic. It does not by itself tell us whether the selected link is legally correct.

Figure 24 checks whether composite score correlates with declared contract duration, which would indicate that short-duration contracts are systematically harder to link (a selection bias concern). No strong correlation is observed, supporting the use of score-based thresholds independently of contract duration.

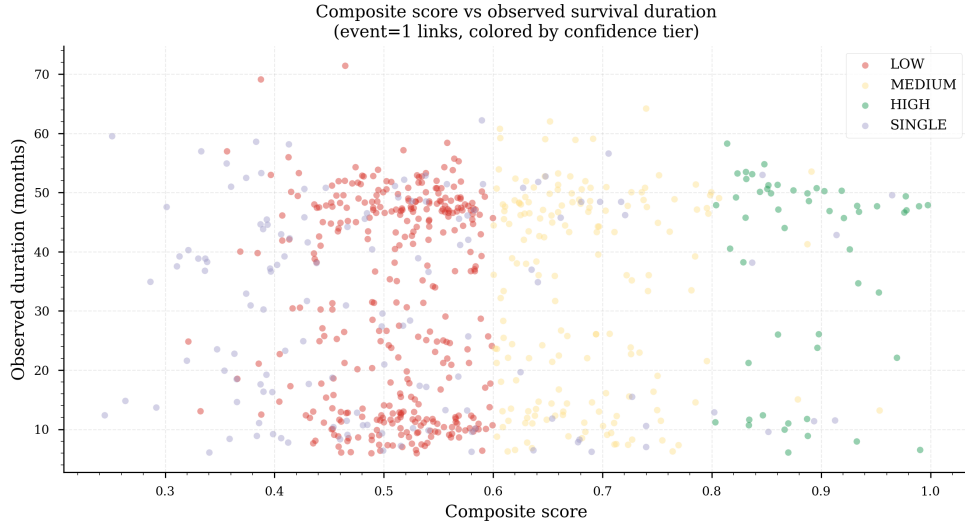


Figure 24: Composite score vs. declared contract duration (months) for the 665 accepted links. No strong duration-linked selection bias is apparent; short contracts (12 months) and long contracts (48+ months) achieve similar score distributions.

4. Placebo / Negative-Control Check

To verify that the scoring function discriminates rather than assigning high scores uniformly, three groups are compared: the winning link (real event), the runner-up (the second-best candidate, not accepted), and a synthetic “no-buyer” candidate constructed by stripping the buyer-identity signal.

The placebo check asks whether the scoring rule gives higher scores to selected links than to plausible but rejected alternatives. If the score had no discriminative value, winning links and runner-up candidates would have similar score distributions.

Table 37: Placebo check: composite score distributions for real winning links, runner-up candidates, and synthetic no-buyer candidates. $\% \geq 0.70$ = fraction reaching the HIGH confidence threshold.

Group	n	Mean	Median	$\% \geq 0.70$
Real winning links (event = 1)	665	0.554	0.529	14.9
Runner-up (rejected candidate)	2,536	0.428	0.420	0.7
Synthetic no-buyer	665	0.401	0.372	4.8

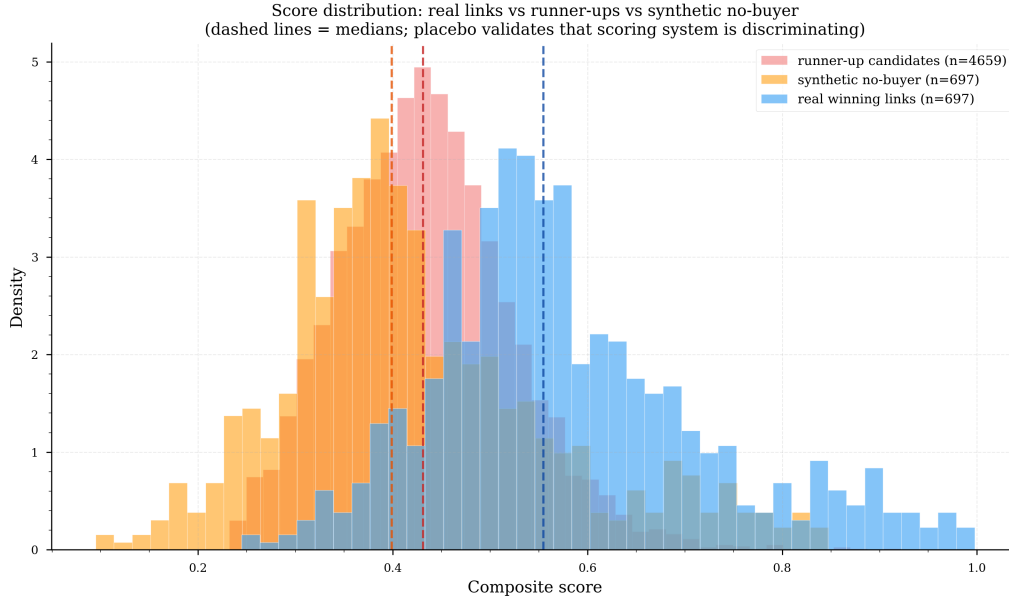


Figure 25: Score distribution for real winning links (green), runner-up candidates (orange), and synthetic no-buyer candidates (red). The clear separation — winner mean 0.554 vs. runner-up 0.428 — confirms that the composite score discriminates meaningfully. Only 0.7% of runner-ups exceed the 0.70 HIGH threshold, compared to 14.9% of winners.

The gap between winners (0.554) and runner-ups (0.428) represents the net discriminative contribution of the four scoring components. The synthetic no-buyer group (0.401) is slightly below runner-ups because some runner-up candidates share the same buyer and thus retain the buyer-identity signal; the small difference (0.428 vs. 0.401) confirms the buyer component contributes positively but is not the dominant signal (the text and temporal components explain most of the winner/runner-up gap).

How to read the result. Winners have a mean score around 0.554, runner-ups around 0.428, and synthetic no-buyer candidates around 0.401. Only 0.7% of runner-ups exceed the 0.70 high-confidence threshold, compared with 14.9% of winners. This is evidence that the score is internally discriminative.

Limitation. This is internal validation, not external ground truth, because the same score is used to select the winning link.

5. Event / Censoring Bias Check

An important validity concern is whether the contracts that received `event = 1` differ systematically from censored contracts on the covariates used in the survival model. A systematic difference could introduce bias into the Cox hazard ratios if the censoring is informative.

For a continuous variable, the standardised mean difference is

$$\text{SMD} = \frac{\bar{x}_{\text{event}=1} - \bar{x}_{\text{event}=0}}{s_{\text{pooled}}}.$$

For binary variables, the table reports the raw difference in proportions. SMD measures imbalance size, not statistical significance. Values below 0.20 are often treated as small in applied diagnostics.

Table 38: Covariate balance between event-1 (linked) and event-0 (censored) contracts. SMD = standardised mean difference; diff. = difference in proportions for binary variables. SMD < 0.20 is conventionally considered small.

Variable	event = 1 (n = 665)	event = 0 (n = 545)	SMD / diff.
Declared duration (months)	33.0 ± 17.1	30.0 ± 18.5	0.166
Contract start year	2018.2 ± 2.1	2018.5 ± 2.3	-0.150
Duration imputed	23.9%	25.0%	-0.010
Amount non-missing	17.1%	25.5%	-0.084
CPV code present	96.5%	93.9%	0.026
Buyer identified by SIRET	0.0%	0.4%	-0.004
N candidates considered	4.8 ± 4.5	0.0 ± 0.0	1.506 (structural)

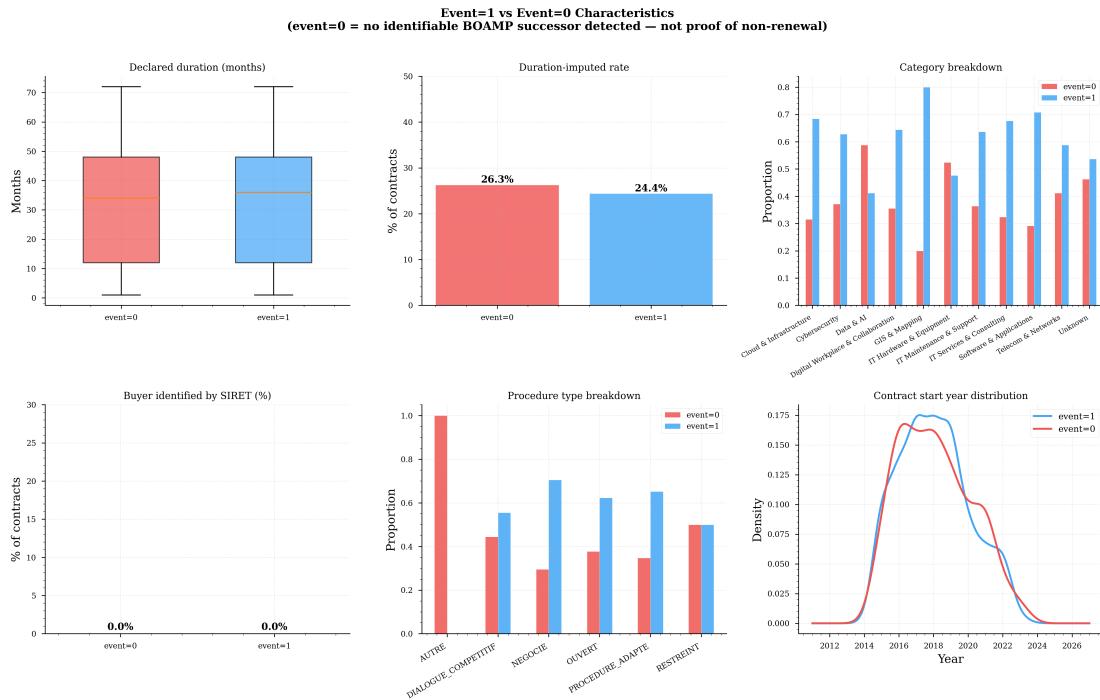


Figure 26: Covariate balance plots comparing event-1 and event-0 contracts. Bars show the standardised mean difference (numeric variables) or difference in proportions (binary variables). All modeling covariates have SMD < 0.20, indicating good balance. The large SMD for N.candidates (1.51) is structural and expected: censored contracts have no temporal partner by definition, so zero candidates were evaluated.

Interpretation. All covariates used in the survival model (declared duration, start year, imputation flag) have SMD below 0.20, indicating that the linked and censored groups are reasonably balanced on these dimensions. This reduces concern about obvious selection bias on the modeled covariates, though it cannot rule out hidden bias or event-label error. The one notable imbalance is amount non-missing (diff. = -0.084): censored contracts are more likely to have a reported amount, suggesting that amount-bearing contracts are slightly harder to re-link — possibly because high-value contracts are more likely to be renegotiated without a new BOAMP call. This does not affect the survival model (amount is not a covariate) but is worth monitoring in future analyses. The large difference for N_candidates is structural and should not be interpreted as a modeling imbalance: censored contracts have no matched temporal partner by construction. The dominant source of censoring is structural right-censoring of the 2022–2024 cohort, whose contracts’ expected renewal windows fall after the 2024-12-31 study close.

14.3 Summary of Phase 2 Findings

- F1. Global median renewal time: 50.1 months.** The KM estimate is consistent with the dominant 48-month administrative duration pattern in public procurement renewal. The distribution is right-skewed: roughly 21% of contracts show an identifiable renewal within 24 months.
- F2. Two statistically significant predictors.** In the multivariate Cox model, only `declared_duration_months` (HR = 0.991, $p < 0.001$) and `start_year` (HR = 1.071, $p < 0.001$) are significant. Shorter declared durations predict faster renewal. The positive `start_year` hazard ratio should be interpreted cautiously: it may reflect calendar-period effects, publication practices, and study-window truncation (2021–2024 contracts are largely right-censored) rather than a genuine acceleration of procurement renewal cycles.
- F3. At most a marginal category effect; no procedure effect.** In the multivariate model, IT Services & Consulting shows a positive effect versus the Cybersecurity reference (HR 1.230, $p = 0.037$); all other category and procedure terms are non-significant. The category log-rank test is likewise marginal ($p = 0.025$). This is a weak signal, not a robust segment effect.
- F4. Log-normal AFT model preferred.** Among parametric models, the log-normal achieves the best trade-off between fit (AIC = 7068.4) and concordance (0.655), outperforming the Cox model marginally. The Weibull model is the worst fit.
- F5. Sensitivity check supports qualitative stability.** Restricting to medium- and high-confidence links (Model B) lifts the KM curve so the median is no longer reached, while leaving covariate effects qualitatively unchanged. Model C is far more aggressive: censoring all but the 75 strict high-confidence links collapses the event rate to 6.2% and

the KM median is no longer reached. The primary model (Model A) is retained as the official result.

The statistical evidence above is mostly internal: threshold sensitivity, score-margin diagnostics, placebo checks, and covariate balance show stability and discriminative structure, but they do not estimate the true precision or recall of the event variable. The manual audit below does, and its verdict is considerably harsher than the internal checks suggest.

6. Manual Validation Audit (executed 2026-07-02)

A stratified sample of 150 cases (100 linked pairs oversampling the strict, ambiguous-margin and single-candidate strata; 50 unlinked sources) was manually audited against the full official BOAMP notice records (verbatim API records; workbook `event_validation/outputs/boa mp_event_validation_audit.xlsx`, labels in `manual_validation_audit_labeled.csv`). For each unlinked source, an active counter-search for missed renewals ran over the entire 3,181-notice corpus (same and fuzzy-similar buyer names, Sentence-Transformer ranking, no window restriction). Wording differences alone were never grounds for rejection.

Table 39: Manual audit results (150 stratified cases). These are sample-based audit estimates, not population-level ground truth.

Quantity	Value
Linked pairs audited (TP / FP / uncertain)	100 (14 / 82 / 4)
Unlinked sources audited (TN / FN / uncertain)	50 (41 / 6 / 3)
Precision, raw sample (decided cases)	0.146
Precision, stratum-weighted to the 665 events	≈ 0.088
Precision, HIGH tier ($S \geq 0.70$) / strict flag	0.50 / 0.50
Precision, MEDIUM tier / LOW tier	0.09 / 0.00
Precision, text similarity ≥ 0.80	1.00 (11/11)
Precision, CPV exact match / weak CPV	0.41 / 0.03
Missed-renewal rate among decided unlinked	0.128 (≈ 70 of 545 censored)
Proxy recall (population-scaled, approximate)	≈ 0.46
Uncertainty rate	4.7% (7/150)

Main false-positive causes: the best same-buyer candidate near the estimated end date is usually a *different* procurement need for large buyers; generic CPV codes (48000000, 72000000) credit unrelated pairs with a perfect CPV component; below text similarity ≈ 0.5 no audited link was a credible renewal; two buyer-key over-merges linked distinct legal entities (conseil régional vs. préfecture). **Main false-negative causes:** early renewals published before the ± 6 -month window opens (administrative-ceiling durations), buyer renames and restructurings, and annual cycles missed by weeks.

Consequence. The proxy event is *not* validated at the baseline definition: the audited precision of the full 665-event set is roughly 9–15%, and the Model A results should be read as describing the timing of algorithm-identifiable same-buyer re-publications rather than validated renewals. Only the high-text-similarity core behaves like a true renewal signal. Recall is likewise imperfect ($\approx 13\%$ of audited censored contracts hide a plausible missed renewal), so the two error types do not cancel. Recommended re-linking changes: raise the text floor to ≥ 0.5 – 0.65 , stop scoring generic CPV codes as exact matches, and widen or de-anchor the temporal window from ceiling durations, then re-estimate the survival models and re-audit.

15 Operational Risk Indicators

Event-definition note (2026-07-05). All indicators in this section are computed from the *pre-calibration* baseline event definition (665 events, log-normal AFT, AIC 7,068.4). They were subsequently re-scored under the calibrated balanced rule (269 events) via `scripts/task_section16b_calibrated_risk_indicators.py` (same design: LogNormal AFT, same covariates, prediction date 2024-12-31). Under the balanced rule the probabilities are substantially lower: mean $\hat{p}_{12} = 0.023$ (vs. 0.074), mean $\hat{p}_{24} = 0.068$ (vs. 0.216), max $\hat{p}_{12} = 0.073$, 27.5 expected recurrences within 12 months (vs. 89.7) and 82.3 within 24 months; the buyer and segment rankings keep the same leaders (SIREN:234400034; IT Services & Consulting). Calibrated outputs: `reports/tables/survival/*_calibrated_balanced.csv`. The subsections below document the pre-calibration run.

15.1 Operational 12/24-Month Renewal Risk Prediction

Business purpose. The preceding survival analysis estimates the *historical* timing of renewals: how long, on average, a digital-sector public contract takes before an identifiable BOAMP renewal signal appears. This subsection translates the fitted survival model into *operational* risk indicators for Gigalis. For each contract still active at a prediction date, it estimates the probability that the contract generates an identifiable BOAMP renewal signal within the next 12 or 24 months, and aggregates these probabilities into buyer-level and segment-level rankings to prioritise renewal-campaign effort.

Prediction model. The **log-normal AFT** model of Section 12 is used as the main prediction model, because it has the best (lowest) AIC among the parametric survival models (AIC = 7068.4; Table 32). It is applied with the same covariates and design matrix as the survival analysis. The prediction date is the study end date, **2024-12-31**; the contract age at that date is $a_i = (2024-12-31 - \text{start_date}_i)$ expressed in months. Predictions are produced for the 1,204 contracts in the model design matrix (the 6 contracts with missing `type_procedure` are excluded, as in the Cox/AFT models).

Conditional renewal probability. For a contract i with current age a_i , covariates X_i , and prediction horizon $h \in \{12, 24\}$ months, the conditional probability of an identifiable renewal within the horizon, given that the contract has not yet shown one, is

$$\hat{p}_{i,h} = \Pr(T_i \leq a_i + h \mid T_i > a_i, X_i) = \frac{\hat{S}(a_i \mid X_i) - \hat{S}(a_i + h \mid X_i)}{\hat{S}(a_i \mid X_i)},$$

where T_i is the time to an identifiable BOAMP renewal, a_i is the contract age at the prediction date, h is the prediction horizon (12 or 24 months), and $\hat{S}(t \mid X_i)$ is the survival probability predicted by the selected log-normal AFT model. By construction $\hat{p}_{i,24} \geq \hat{p}_{i,12}$ (the survival function is monotone non-increasing); probabilities are clipped to $[0, 1]$ and the 0/0 case ($\hat{S}(a_i) \approx 0$) is set to zero. Across the 1,204 contracts the 12-month probability ranges from 0.003 to 0.288 (mean 0.075) and the 24-month probability from 0.023 to 0.564 (mean 0.216); the expected number of identifiable renewals is 89.7 within 12 months and 259.9 within 24 months. (These are markedly lower than at the earlier $W = 12$ setting, where the expected 12-month count was 208: the narrower $W = 6$ window shifts the fitted model toward lower conditional renewal probabilities.)

How to read the result. This probability is conditional. It does not ask, “Will the contract renew within 12 months from its start date?” It asks, “Given that this contract has reached its current age without an observed renewal, what is the probability that an identifiable BOAMP renewal appears within the next 12 or 24 months?”

At the buyer level, the expected number of renewals is the sum of the contract-level probabilities:

$$E[\text{renewals}_{12\text{m}} \text{ for buyer } b] = \sum_{i \in b} \hat{p}_{i,12}.$$

This is why a buyer with many moderate-risk contracts can rank above a buyer with one high-risk contract.

Risk tiers. Contracts are assigned an operational risk tier from $\hat{p}_{i,12}$: **High** (≥ 0.50), **Medium** ($0.25 \leq p < 0.50$), and **Low** ($p < 0.25$). On the current data this yields 0 High, 6 Medium, and 1,198 Low contracts: because most contracts are already several years old, no contract reaches a 12-month conditional probability of 0.50. These are *operational* thresholds, to be calibrated later with the supervisor and with the manual-validation sample.

Output files. Three operational tables are produced (Table 40).

Table 40: Operational prediction output files (in `reports/tables/survival/`)

File	Rows	Purpose / main columns
<code>renewal_risk_12_24_months.csv</code>	1,204	Contract-level predictions: one row per scored contract. Main columns: <code>contract_id</code> , <code>buyer_key</code> , <code>category_label</code> , <code>age_months</code> , <code>p_renewal_12m</code> , <code>p_renewal_24m</code> , <code>risk_tier</code> , <code>model_used</code> .
<code>buyer_renewal_risk_ranking.csv</code>	321	Buyer-level ranking by expected renewals. Main columns: <code>buyer_key</code> , <code>n_contracts</code> , <code>expected_renewals_12m/_24m</code> , <code>max/mean_p_renewal_12m</code> , <code>high_risk_contracts_count</code> .
<code>segment_renewal_risk_ranking.csv</code>	11	Segment-level ranking by expected renewals. Main columns: <code>category_label</code> , <code>n_contracts</code> , <code>expected_renewals_12m/_24m</code> , <code>mean_p_renewal_12m/_24m</code> , <code>high_risk_contracts_count</code> .

Top contracts. Table 41 lists the 10 contracts with the highest 12-month renewal-risk probability.

Table 41: Top 10 contracts by $\hat{p}_{i,12}$ (12-month renewal-risk probability)

Contract ID	Buyer (normalised key)	Segment	p_{12}	p_{24}
BOAMP:24-12133	tribunal administratif de nantes	IT Services & Consulting	0.288	0.565
BOAMP:24-21999	synd. dép. d'énergie (SYDEV)	IT Services & Consulting	0.282	0.557
BOAMP:24-20387	cons. dép. loire-atlantique	Software & Applications	0.273	0.547
BOAMP:22-166149	gie des epl de vendée	Software & Applications	0.267	0.540
BOAMP:23-98097	vendée grand littoral	Software & Applications	0.261	0.532
BOAMP:22-112462	semitan	IT Services & Consulting	0.251	0.520
BOAMP:22-153148	cons. rég. pays de la loire	IT Services & Consulting	0.239	0.504
BOAMP:23-131022	rectorat acad. de nantes	Software & Applications	0.236	0.501
BOAMP:20-100172	comm. urbaine d'alençon	IT Services & Consulting	0.231	0.494
BOAMP:24-7288	département de la vendée	Cybersecurity	0.230	0.492

Top buyers. Table 42 ranks buyers by expected 12-month renewal volume.

Table 42: Top 10 buyers by expected identifiable renewals in 12 months ($\sum_i \hat{p}_{i,12}$)

Buyer (normalised key)	n	$E[\text{ren.}]_{12}$	$E[\text{ren.}]_{24}$	mean p_{12}
cons. rég. pays de la loire	91	9.34	25.54	0.103
nantes métropole	72	3.39	11.51	0.047
cons. dép. loire-atlantique	28	3.24	8.53	0.116
chu de nantes	42	2.61	7.83	0.062
angers loire métropole	30	2.34	6.80	0.078
ch le mans	22	2.24	5.98	0.102
commune de saint nazaire	35	2.23	6.92	0.064
gie sesam vitale	28	2.10	6.32	0.075
département de la vendée	19	1.93	5.14	0.101
vendée expansion	9	1.39	3.43	0.155

Segments. Table 43 gives the expected 12- and 24-month renewal volume by service segment.

Table 43: Expected identifiable renewals by service segment (12 and 24 months)

Segment	n	$E[\text{ren.}]_{12}$	$E[\text{ren.}]_{24}$	mean p_{12}	mean p_{24}
IT Services & Consulting	343	30.36	84.14	0.089	0.245
Telecom & Networks	220	15.90	46.14	0.072	0.210
Software & Applications	223	14.57	43.77	0.065	0.196
Cybersecurity	136	10.18	29.68	0.075	0.218
Unknown	114	7.63	22.97	0.067	0.201
Digital Workplace & Collaboration	51	2.89	9.02	0.057	0.177
Data & AI	40	2.32	7.22	0.058	0.180
Cloud & Infrastructure	22	1.75	5.04	0.080	0.229
IT Maintenance & Support	22	1.61	4.81	0.073	0.219
IT Hardware & Equipment	25	1.58	4.80	0.063	0.192
GIS & Mapping	8	0.89	2.32	0.112	0.290

Distributions and rankings. Figure 27 shows the distribution of the 12- and 24-month renewal-risk probabilities; Figure 28 shows the top buyers and the segment-level expected renewal volume.

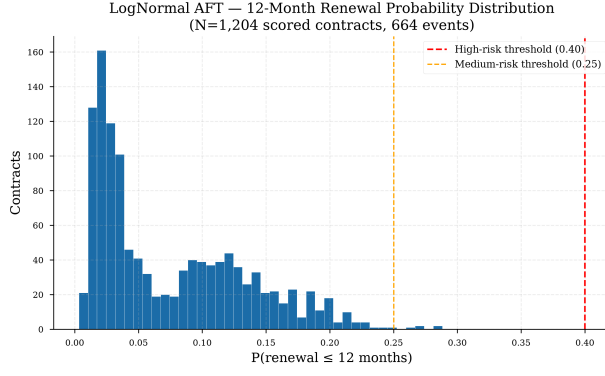
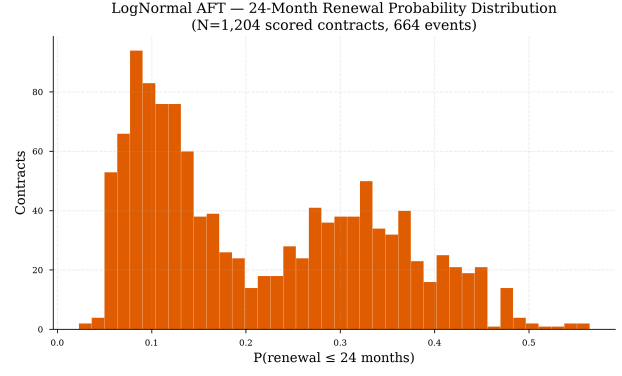
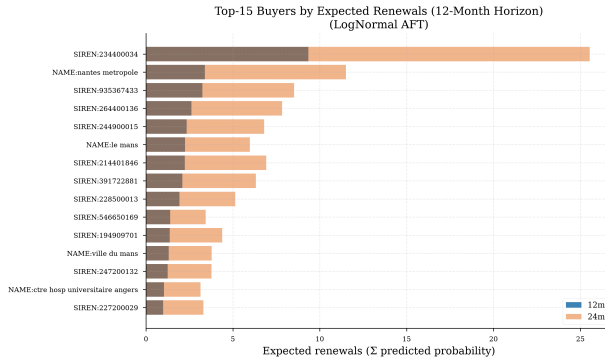
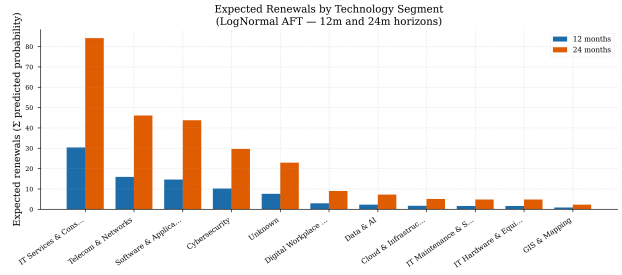
(a) 12-month probability $\hat{p}_{i,12}$ (b) 24-month probability $\hat{p}_{i,24}$

Figure 27: Distribution of contract-level renewal-risk probabilities under the log-normal AFT model (proxy-linking rule, $N = 1,204$).

(a) Top 15 buyers by $\sum_i \hat{p}_{i,12}$ 

(b) Expected renewals by segment (12 vs 24 mo)

Figure 28: Operational renewal-risk rankings: buyers with the largest near-term expected renewal volume (left) and expected renewal volume by service segment (right).

Limitation. These probabilities estimate the risk of an identifiable BOAMP renewal under the current proxy-linking rule. They should be interpreted as prioritization indicators, not as legally verified renewal probabilities. The manual audit (Section 14.3) estimates the precision of the proxy event at only $\approx 9\text{--}15\%$, so these indicators inherit that uncertainty and should be recalibrated after the recommended re-linking.

16 Reproducibility and Output Files

All results in this report are reproducible from tracked outputs. The survival models are rebuilt by `notebooks/02_survival_modeling_boamp.ipynb` (generated by `_build_survival_notebook.py`); the robustness checks by `validation_robustness/`. The calibrated event-definition workflow (calibration update at the front of this report) is reproduced by executing notebooks `04_synthetic_boamp_benchmark.ipynb`, `05_parameter_calibration_benchmark_and_real.ipynb`, `06_apply_calibrated_rules_to_real_boamp`, and `07_calibrated_survival_analysis.ipynb` in order; its key outputs are `reports/tables/validation`, the three `boamp_phase2_survival_calibrated_*.csv` datasets (1,210 rows each), and `reports/tables/survival/calibrated_rule_{km_summary,cox_comparison}.csv`. Survival figures are written to `reports/figures/survival/` and result tables to `reports/tables/survival/`. The Phase-1 linking workflow produces five output files:

Table 44: Output files produced by the BOAMP linking workflow

File	Rows	Purpose
<code>boamp_renewal_links.csv</code>	1,210	Notebook linking output. One row per eligible AO before final handoff standardisation.
<code>boamp_phase2_survival_method_m0_balanced.csv</code>	1,210	Current modeling input. BOAMP-only handoff under the selected M0 balanced proxy-event definition.
<code>boamp_phase2_survival.csv</code>	1,210	Historical pre-calibration 665-event handoff retained for provenance and comparison.
<code>boamp_renewal_candidates.csv</code>	3,201	All filtered candidate pairs before best-match selection. Useful for threshold sensitivity analysis.
<code>boamp_linking_stats.csv</code>	1	Summary statistics: total AO, eligible, linked, linking rate.
<code>boamp_bias_report.csv</code>	67	Failure reason $\times$ CPV cross-tabulation. Diagnoses which categories are under-linked.

17 Remaining Work

Phase 2, including the operational renewal-risk prediction layer (Section 15.1), is complete. The remaining planned steps are:

1. **Manual validation — executed 2026-07-02 (Section 14.3):** 150 stratified cases audited; precision ≈ 0.09 – 0.15 overall (0.50 HIGH/strict tier, 1.00 at text ≥ 0.80 , 0.00 LOW tier); missed-renewal rate 12.8% among audited censored contracts. The follow-up re-linking under stricter rules was executed as the 2026-07-05 calibration (notebooks 04–07).
2. **Event-definition calibration — executed 2026-07-05:** broad / balanced / strict rules selected from the synthetic benchmark (scored with the real Sentence-Transformer encoder) and real diagnostics; survival rerun completed under all three rules, and the operational risk indicators were re-scored under the balanced 269-event rule (`task_section16b_calibrated_risk_indicators.py`).
3. **Trend / change-point analysis:** Analyse renewal-rate trends and structural breaks by technological segment.
4. **Optional DECP enrichment:** Revisit DECP integration only after the BOAMP-only model is stable and a defensible cross-source linkage strategy is defined. Exploratory artefacts already exist (`decp_renewal_links.csv`, `buyer_bridge.csv`, `unified_survival.csv`) but remain outside the validated BOAMP-only handoff.

L3 status. The internship guide’s L3 deliverable (survival analysis notebook: KM curves, validated Cox model, Weibull/parametric comparison, 12/24-month individual predictions, high-risk contract table) is **Done** on the official M0 balanced dataset — see the Cox PH assumption check above (Section), the AFT comparison including Weibull (not retained; LogNormalAFT selected), and the risk-ranking tables in Section 15. The trained NLP classifier (L2) is a separate team workstream and does not block L3: the survival analysis uses the rule-based CPV/keyword taxonomy (`data/processed/taxonomy.csv`, Section 5.2) as a provisional segment covariate, and the pipeline can be rerun once final NLP-classifier labels are delivered.

Buyer SIREN enrichment (completed 2026-06-24) — results incorporated. A buyer-name resolution pipeline (`buyer_siren_enrichment/`, 7 steps) was built and executed after the initial linking. It queries the API Recherche d’Entreprises to map each unique `nomacheteur` string to a SIREN identifier, enabling deduplication of buyer aliases that name normalisation cannot resolve. Results: 275 of 525 unique buyers (52.4%) matched with HIGH confidence; 23 previously fragmented buyer groups were merged. On the **auxiliary Jaccard comparison** (all 1,933 AO; output files: `boamp_full_survival.csv` name-based baseline and `boamp_full_survival_enriched.csv` in `boamp_renewal_linking_siren_enriched/outputs/`, kept as **legacy comparison artifacts** to evaluate the Jaccard method only — not the current-stage survival dataset), the event rate rose from 7.6% to 8.3% at the current $W = 6$ setting (+15 events; log-rank $p = 0.38$, not significant; at the earlier $W = 12$ setting the comparison was 11.3% to 12.1%). **The sentence-transformer pipeline was then fully re-run** with `buyer_key_enriched` as the active grouping key. The

23 SIREN merges expanded the candidate pool for affected source contracts, producing **665 pre-calibration proxy events** (W=6 rerun; pre-W=6: 705 events at W=12). The later 2026-07-05 calibration selects the balanced 269-event rule as the recommended main survival input. Key impacts on the survival analysis:

- W=6 rerun (2026-07-02): linking rate 55.0% (665/1,210); previously 64.1% (705/1,100) at W=12.
- SIREN merges expanded the candidate pool; Model B (composite ≥ 0.50) counts 400 events (W=6 run; pre-enrichment comparison not separately measured).
- Strict H.C. (Model C): 75 events (score ≥ 0.70 , margin ≥ 0.05).
- This was superseded for reporting by the 2026-07-08 method-comparison rerun: selected M0 balanced has KM median not reached, Cox C-index 0.544, and LogNormalAFT AIC 3,542.5.
- The pre-enrichment limitation that most buyer keys were name-based has been materially reduced. In the current enriched survival population, 724 rows are SIREN-keyed (plus 2 SIRET) and 484 remain NAME-keyed; residual buyer fragmentation now concerns only the unresolved NAME-keyed buyers.

Additionally, the official phase-2 dataset has been **annotated** with buyer enrichment metadata in `data/processed/boamp_phase2_survival_siren_enriched.csv` (1,210 rows, four added columns: `buyer_key_enriched`, `enriched_siren`, `enrichment_confidence`, `enrichment_score`, added by left-join; the event and survival-time columns are unchanged). Enrichment-confidence breakdown over the 1,210 phase-2 rows: **HIGH 724 (59.8%)**, MEDIUM 91, LOW 49, NO_MATCH 346. The 484 rows without a HIGH-confidence match retain the original name-based buyer key. Note that HIGH-confidence automated matching is not equivalent to manual validation; rare false positives at HIGH confidence remain possible.

All figures and tables in this report reflect the enriched re-run. The full buyer-resolution method is documented in Section 5.1, and the CPV/taxonomy method is documented in Section 5.2.